When Selling Spreads: Local Decisions, Limited Resources, and Market Crashes in a Simulated Market*

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Abstract

Publicly expressed trading intentions can influence others even when no transaction occurs. We developed a market model in which traders on a grid responded to neighbors' intentions without observing their holdings, while technical traders responded to price. Paired markets shared a history before selected traders in one market received a lasting change in their decision rule. Under the original observation rule, a Sell majority formed once enough traders received the new rule, even when technical traders were kept in Hold and no decision responded to price. Majority probability rose over a narrow range of switched fractions that moved with market size, observation time, and revision frequency. Orders cleared against a finite constant-product dealer. Shallow positive reserves allowed severe declines even without a rule switch, so its effect was measured against paired controls. Giving traders fewer shares reduced declines, while transaction fees reduced them in some conditions and deepened them in others. In a hypothetical information regime, traders read Hold when a neighbor intended to sell but had no shares. This change reduced the switch's effect on late selling intentions and drawdown in the primary comparison, while majorities reappeared at higher switched fractions. In separate experiments, traders following one shared source of random advice spread selling and, with deep dealer reserves, gained cash at other traders' expense. The model connected the spread of public intentions to their financial consequences and showed how those consequences depended on the observation and execution rules.

*Repository: https://github.com/varnerlab/Wolfram-markets-abm-paper.git.

Preprint.

1 Introduction

Financial markets are the collective outcome of human interactions. They reflect both our ingenuity and our failings. In its 2011 report on the 2008 financial crisis, the Financial Crisis Inquiry Commission paraphrased Shakespeare: "the fault lies not in the stars, but in us" [1, p. xvii]. In markets for traded financial assets such as stocks, investors make decisions with different information, expectations, and financial resources, and they respond to prices and to the actions of others when deciding whether to buy, hold, or sell. An investor's decision to sell can prompt others to do the same. Sellers need buyers who are willing and able to purchase their assets, and the prices buyers will pay reflect their views of the assets' value and the funds they can commit to the purchase. If sellers cannot find buyers at the current price, they may accept lower prices to complete their sales, and the resulting price changes can then influence investors' later decisions. Under what conditions can a change in the behavior of some investors lead to widespread selling, and when does that selling produce a market crash?

Agent-based models study how individual trading decisions and interactions among investors produce market behavior. This literature includes many approaches to learning, imitation, and price formation [2, 3], and its models differ in the information available to traders and in how their trades affect prices. Farmer and Joshi studied traders who followed price trends or compared prices with their estimates of asset value. A market maker absorbed their net orders and adjusted the price, and the resulting price changes then influenced later trading decisions [4]. In the Santa Fe artificial stock market, traders learned forecasting rules from market prices and dividends and used those forecasts to guide their trades [5]. In previous work, we developed an agent-based market for trading multiple assets, in which agents submitted orders to a continuous double auction that matched compatible buy and sell orders as they arrived. Traders with different decision rules therefore interacted through a common trading mechanism [6]. Alongside interactions through prices and trades, herding models describe how traders respond to the behavior of others [7–9]. Arranging traders in rows and columns provides a simple representation of the connections through which influence operates: each position represents a trader, and neighboring positions identify the traders whose decisions it can observe. The information passed through these connections can be as simple as a trader's current decision to buy or sell [10], and a change in one trader's decision can affect its neighbors, whose responses can then influence other traders. Models inspired by the Ising model use these local interactions to represent imitation. In Bornholdt's simplified model, traders tended to follow their neighbors, but a growing imbalance between buyers and sellers across the market increased their tendency to reverse their current decisions [11]. Iori used a related model in which traders could buy, sell, or remain inactive, repeatedly exchanged their current decisions with their neighbors, and used the resulting signals to revise their own choices [10]. These models used similar arrangements of traders but different rules for responding to neighboring decisions. The arrangement specifies who can influence whom, while the decision rule specifies how each trader responds.

Cellular automata use local rules to update traders' decisions in discrete steps [12, 13]. Wolfram arranged traders on a line, where each trader responded to its two neighbors and the cumulative difference between buying and selling served as a price analogue [14]. Later models combined cellular automata with trader balance sheets [15] or used stochastic rules for interactions between neighboring traders [16–20]. Our model represents a synthetic market with traders arranged in rows and columns. Two types of traders occupy the market interior, and both use their four neighbors' intended actions to decide whether they want to buy, hold, or sell. Their rules differ in a single response to their neighbors' intended actions, and we vary the relative numbers of the two types to study how the mixture affects the spread of selling intentions. Technical traders along the edge of the synthetic market base their decisions on prices, and the interior traders next to them respond to those decisions. The financial consequences of intended trades depend on the resources available for exchanging cash and shares, and earlier agent-based models have examined limits on traders' cash and asset holdings and on their ability to borrow [21, 22]. In our model, Buy and Sell indicate publicly communicated intentions, whereas a completed trade transfers shares from a seller to a buyer and cash from the buyer to the seller. Neighboring traders observe these intentions without observing holdings or verifying transactions. This assumption represents influence through statements about intended trading, whose execution an observer cannot check. Traders cannot borrow cash or sell shares they do not own. A trader without shares can still indicate a desire to sell but cannot carry out the sale, and a trader without enough cash can indicate a desire to buy but cannot carry out the purchase. Because neighboring traders respond to intended actions even when no trade occurs, selling

intentions can spread without a corresponding increase in completed sales. A dealer acts as the market maker, using its finite cash and share reserves to complete trades that cannot be matched between traders. Completed trades preserve the product of the dealer's cash and share reserves, following a constant-product market-making approach [23], and after each round of trading the dealer's quoted price per share equals its remaining trading cash divided by the number of shares it holds. When the dealer buys shares from traders, it has less cash and more shares, so dividing the smaller cash balance by the larger number of shares gives a lower quoted price. The size of the dealer's reserves therefore determines both which trades can be completed and how much the quoted price changes when the dealer buys or sells shares. Technical traders use the dealer's updated quoted price to make their next decisions, which then influence neighboring interior traders.

In this study, we constructed a synthetic market based on a three-state cellular automaton to examine whether mixing two trader types could lead to widespread selling intentions, and how trader holdings and dealer reserves affected the accompanying price decline. In each paired rule-switch comparison, both markets had the same number of traders and shared a period of trading with only the original trader type in the interior. We then assigned the second decision rule to a chosen fraction of interior traders in one market, while the other market kept the original rule. Trading intentions, trader holdings, dealer reserves, and prices matched at the rule switch. The traders could then make different decisions and complete different trades. We varied the fraction of interior traders using the second rule to determine when at least half of the interior traders wanted to sell, which we called a Sell majority. In separate experiments, we varied the total number of traders and the simulation duration to examine how market size and time affected the spread of selling. We also ran markets with all technical traders held in Hold to test whether a Sell majority could develop when no decision responded to the price. Interior traders continued to respond to neighboring intentions and could complete trades with one another or the dealer, so trades with the dealer could still change its reserves and quoted price. In the rule-switch experiments, keeping technical traders in Hold prevented price changes from influencing interior traders' intentions, but prices still affected which trades could be completed. Buyers needed enough cash to purchase shares, sellers needed shares to sell, and each sale required a buyer or dealer able to pay the execution price. Because every trader in the main model revised its decision at every step and technical traders stood only along the edge, we also ran markets in which traders revised their decisions less often and markets with technical traders inside the grid. We varied dealer reserves and traders' cash and shares separately to examine how those resources affected completed trades and price declines. In a hypothetical information regime, we also let traders read Hold in place of a neighbor's Sell intention when that neighbor had no shares, to test how the spread of selling depended on accepting signals without checking their backing. We also examined crowding by having groups of traders follow shared advice or copy another trader's completed trades. We varied the fraction of followers and the number of sources to examine effects on other traders' decisions, prices, and the distribution of gains and losses. For shared advice, we tested whether followers' influence on neighboring traders contributed to their financial gains. Together, the rule-switch and following experiments showed how individual trading decisions could spread through a market, affect prices, and redistribute cash among traders.

2 Methods

2.1 Three-state cellular automaton

We represented the synthetic market as L rows and L columns, with $(L-2)^2$ traders in the interior, and recorded each trader's intended action as Buy ($s = 0$), Hold ($s = 1$), or Sell ($s = 2$). Technical traders occupied selected positions along the edge and formed their intentions from the dealer's quoted price and recent price history, while the remaining edge positions stayed in Hold. Each interior trader formed its local decision from the intentions of its four neighbors above, below, to the left, and to the right. Each trader's next intention depended only on the sum of these four neighboring states and not on its own current state, so for interior traders next to the edge this sum could include the intentions of technical traders. In the terms of Packard and Wolfram, whose totalistic rules also add the cell's own state to the sum, these are outer totalistic rules that ignore the cell's own state [13]. After trades settled at each step t, the technical traders updated their intentions using the new quoted price and price history. All interior traders then updated together using the previous interior states and the newly updated edge states, producing intentions for the next step's orders. We denoted the four

neighbors of trader i by ∂i and the state supplied by neighbor j at step t by $s_j^{\text{in}}(t)$, and the decision rule used their sum $c_i(t)$, defined by:

$$c_i(t) = \sum_{j \in \partial i} s_j^{\text{in}}(t) \in \{0, 1, \ldots, 8\}. \tag{1}$$

Each decision rule assigned an intended action to each of the nine possible neighbor sums. For a sum $k \in \{0, 1, \ldots, 8\}$, we denoted the trader's next intended action by $a_k \in \{0, 1, 2\}$, using the codes for Buy, Hold, and Sell. Specifying these nine outputs defined the complete rule, giving $3^9 = 19{,}683$ possible rules for the four-neighbor model. We identified each rule by its Wolfram rule number R, with the outputs written as base-three digits from a_8 on the left to a_0 on the right [12, 13], so the rule number was defined by:

$$R = \sum_{k=0}^{8} a_k 3^k. \tag{2}$$

We used Rule 19280, $(222110002)_3$, as the baseline and Rule 19286, $(222110022)_3$, for the selected traders after the rule switch. The two rules differed only in their response to neighbor sum $k = 1$, corresponding to three neighbors intending to Buy and one intending to Hold. For this configuration Rule 19280 assigned Buy and Rule 19286 assigned Sell as the trader's next intended action, whereas all other neighbor sums produced the same intended action under both rules. Before each run, we specified an approximate fraction f of interior traders to receive Rule 19286 later in the simulation and selected $\text{round}(f(L-2)^2)$ traders uniformly without replacement. We initialized the interior with $\text{round}(0.05(L-2)^2)$ traders intending to Buy and the same number intending to Sell, at positions chosen uniformly without replacement, and all remaining interior traders began in Hold. All interior traders initially used Rule 19280, allowing the market to develop before the scheduled rule switch. At the switch, the selected traders received Rule 19286 and first used it to determine their next intended action, while all other interior traders continued using Rule 19280. Assigning the new rule did not execute a trade or reset any trader's cash or shares.

2.2 Orders, trader resources, and completed trades

A trader's intended action generated a desired order, but completing the order required cash or shares and a counterparty. Each interior trader and technical trader held cash w_i and an integer number of shares q_i. A Buy action requested one share, a Sell action offered one share, and a Hold action generated no order. All completed trades within a step used the same base execution price, and a buy required enough cash to pay the execution price and any transaction fee, whereas a sale required at least one share. We allowed sales below a trader's average purchase price $\bar{c}_i$, so the model did not impose the reluctance to realize losses described in behavioral studies [24, 25]. We classified desired orders that could not be completed because the trader lacked cash or shares as *trader suppression*. In the main model, we interpreted Buy and Sell as publicly communicated intentions. Neighbors read these signals without observing the sender's holdings or checking whether a trade completed, so an intention could influence them even when its sender lacked the resources to act on it. Cash and share holdings therefore limited completed trades but did not determine which intentions entered neighboring decisions. The model did not give traders a separate choice of what to announce, so this interpretation did not represent strategic misreporting.

The dealer acted as market maker for orders unmatched between traders. Clearing fixed the net trade direction, selected its largest feasible size and price, then allocated orders at that price. For N_{tr} interior and technical traders, dealer depth h, the dealer's initial shares per trader, set initial shares $H = \text{round}(hN_{\text{tr}})$ and cash $P_0 H$, with initial price $P_0 = 100$. The dealer could not borrow and, for positive reserves, retained at least one share while preserving the product K of trading cash and shares [23]. At step t, the starting reserves were cash W_{t-1} and shares Q_{t-1}. A proposed net purchase x from the dealer was positive for trader buys and negative for trader sales. Its base execution price $F_t(x)$ had to preserve the reserve product after the dealer exchanged x shares for cash $F_t(x)x$, according to:

$$K = W_{t-1} Q_{t-1},$$
$$[W_{t-1} + F_t(x)x][Q_{t-1} - x] = K.$$

For a nonzero proposed net trade, the corresponding price was:

$$F_t(x) = \frac{W_{t-1}}{Q_{t-1} - x}, \qquad Q_{t-1} - x \geq 1. \tag{3}$$

For no net trade, we used $F_t(0) = W_{t-1}/Q_{t-1}$ to match opposing orders without changing dealer reserves. The fee fraction of each completed order's base price was τ, with $\tau = 0$ except in fee comparisons. At each proposed price $F_t(x)$, the eligible counts were $B_t(x)$ Buy orders backed by cash $(1 + \tau)F_t(x)$ and $S_t(x)$ Sell orders backed by at least one share. First, we fixed the trade direction using eligible orders at $F_t(0)$, the price for no net trade. The direction d_t was positive for net purchases from the dealer, negative for net sales, and zero for no net trade, as given by:

$$d_t = \text{sign}\{B_t(0) - S_t(0)\}.$$

Next, with a nonzero direction fixed, we chose the largest nonnegative integer size k that left the dealer at least one share and was supported by the excess eligible orders at its own proposed price. The selected net purchase x_t was therefore:

$$x_t = d_t \max\left\{k \in \mathbb{Z}_{\geq 0} : Q_{t-1} - d_t k \geq 1, \quad d_t[B_t(d_t k) - S_t(d_t k)] \geq k\right\}. \tag{4}$$

We set $x_t = 0$ when $d_t = 0$. Finally, we allocated orders at the selected price $F_t(x_t)$, fixing the eligible counts at $B_t = B_t(x_t)$ and $S_t = S_t(x_t)$. The completed trader buys n_t^B and sells n_t^S were given by:

$$n_t^B = \min\{B_t, S_t + x_t\}, \qquad n_t^S = n_t^B - x_t. \tag{5}$$

All eligible orders on the smaller side completed. Orders on a rationed side were ranked by a reproducible score using simulation seed, step, trader identity, and side, with identity breaking ties. Eligible orders left unfilled were *dealer rejection*. Rejection could reflect the minimum share reserve or a price at which eligible orders no longer supported a larger net trade, without exhausting dealer cash. The selected net purchase per trader ϕ_t summarized completed orders without separately setting price:

$$\phi_t = \frac{x_t}{N_{\text{tr}}}. \tag{6}$$

All completed orders used the common base execution price $F_t = F_t(x_t)$, so the dealer's remaining shares Q_t and trading cash W_t were:

$$Q_t = Q_{t-1} - x_t, \qquad W_t = W_{t-1} + F_t x_t. \tag{7}$$

After settlement, we set the dealer's quoted price P_t equal to its remaining trading cash divided by its remaining shares, and substituting the base execution price into these balances gave:

$$P_t = \frac{W_t}{Q_t} = P_{t-1}\left(\frac{Q_{t-1}}{Q_{t-1} - x_t}\right)^2. \tag{8}$$

This quoted price supplied the price signal used by technical traders and differed from the base execution price whenever $x_t \neq 0$. For each completed order, a buyer paid $(1 + \tau)F_t$ and a seller received $(1 - \tau)F_t$, and the dealer collected the fees in a separate cash account G_t, initialized at zero and excluded from the trading reserves used to set prices:

$$G_t = G_{t-1} + \tau F_t(n_t^B + n_t^S), \qquad G_0 = 0. \tag{9}$$

No fee was charged on an unfilled order. Cash was conserved across traders, dealer trading reserves, and the fee account, except at scheduled changes in trader resources, and shares were conserved across traders and the dealer, except at scheduled share interventions. Dealer reserves determined both the funding available for trades and the response of the quoted price to net trading. The comparison with large reserves used dealer depth $h = 100$, whereas at dealer depth $h = 0$ the dealer held no trading cash or shares. Only opposing trader orders backed by sufficient resources could then be matched, the price remained at P_0, and fees collected in this case did not fund dealer trading. This separate case therefore provided a fixed-price control (Algorithm S2).

2.3 Technical traders and price feedback

Technical traders brought changes in the dealer's quoted price into neighboring decisions. We selected one quarter of the edge positions uniformly without replacement for these traders, while the remaining edge positions stayed in Hold. The technical traders followed a contrarian strategy of the Bollinger-band type, in which each technical trader compared the quoted price with a band around a recent price average. A price below the lower limit generated a Buy intention, a price above the upper limit generated a Sell intention, and prices within the band, including its limits, generated Hold. These intentions influenced neighboring interior traders and supplied orders for the next trading step. This contrarian response contrasted with the trend-following moving-average rules often tested in market data [26]. The reference price μ_t was an exponentially weighted moving average of the dealer's quoted price, and the band width used σ_t, an exponentially smoothed absolute departure from this average rather than a standard deviation. Both quantities used the smoothing parameter $\alpha = 1/60$. After settlement, we first updated the average using the new quoted price P_t and then updated the departure:

$$\begin{aligned} \mu_t &= \mu_{t-1} + \alpha(P_t - \mu_{t-1}), \\ \sigma_t &= \sigma_{t-1} + \alpha\big(|P_t - \mu_t| - \sigma_{t-1}\big). \end{aligned} \tag{10}$$

We initialized the average at $\mu_0 = P_0$ and the departure at $\sigma_0 = 0$. Each technical trader received a fixed multiplier $k_j \sim U[0.5, 1.5]$, giving lower and upper limits $\mu_t - k_j\sigma_t$ and $\mu_t + k_j\sigma_t$.

At the start of each step, we applied any scheduled changes to decision rules or trader resources, and traders then submitted orders from their existing intentions. After settlement, technical traders updated their bands and intentions using the dealer's new quoted price, and interior traders then updated simultaneously, using the previous intentions of neighboring interior traders and the new intentions of adjacent technical traders. The resulting intentions supplied orders for the following step (Algorithm S1). To remove price feedback, we kept every technical trader in Hold from initialization, so these traders submitted no orders and communicated Hold to their neighbors. Interior traders could still trade, and cash, shareholdings, and the dealer's quoted price could still change, but those changes could no longer influence interior intentions through the technical traders. We compared this condition, *technical traders held*, with *technical traders active*, in which they followed the band rule. Both conditions began with the same starting market and used the same interior traders selected for the rule switch, so the comparison tested whether a Sell majority could develop from responses to neighbors without feedback from prices. Figure legends abbreviate these conditions as held and active.

2.4 Paired rule-switch experiments

We measured rule-switch effects by pairing switched markets with no-switch controls that retained the original rule. Each pair began from the same starting market and used Rule 19280 for every interior trader for 1,000 steps. At the start of step 1,001, we assigned Rule 19286 to a preselected fraction f of interior traders, who retained their current intentions and financial resources, while the control continued with Rule 19280. The two simulations therefore shared the same history up to the rule switch. Each condition in the four main comparisons used 50 starting markets (Supplementary Table S1). Every trader initially held 1,000 cash units and ten shares, and each simulation continued for 3,000 steps after the baseline. Additional comparisons tested other decision rules and a one-time instruction to sell, and we also kept the technical traders' band widths fixed after the rule switch, tested additional dealer depths, and extended the observation period. The supplementary protocols give the conditions and parameter values for these comparisons (Appendix A.5).

We varied the fraction of traders given the new rule, market size, and financial resources to compare the spread of Sell intentions with changes in completed trades and prices. The rule-switch comparison used a 64×64 grid, active technical traders, and dealer depth $h = 100$ at fractions $f \in \{0.10, 0.15, 0.20, 0.25\}$, whereas the dealer-depth comparison used 11 fractions between 0.10 and 0.25 at eight dealer depths h from 0 to 100, with technical traders active or held. For the size comparison, we used grid sizes $L \in \{32, 48, 64, 96, 128\}$, with technical traders held at dealer depth $h = 100$ or active at $h = 1$ and $h = 100$. At a fixed dealer depth h, initial dealer cash and share reserves increased in proportion to the number of traders. To test how trader resources affected the response to the rule switch, we independently multiplied each trader's current cash and shareholdings by factors $g_w, g_q \in \{0.1, 1, 10\}$ at the switch, which gave nine combinations, including unchanged

holdings when both factors were 1. Share counts were rounded to whole numbers, and the intervention did not rescale the dealer's reserves. With other settings fixed, all nine simulations shared the same history before the intervention and assigned the new rule to the same traders. These resource comparisons used switched markets without separate no-switch controls. When technical traders were held, changing resources could alter completed trades and prices but could not alter interior intentions, because neither cash nor shareholdings entered the decision rule. The supplementary material gives the full parameter combinations and verification checks (Appendix A).

We tested fees of 0, 5, 10, and 25 basis points on each completed buy or sell to determine whether trading costs changed the response to the rule switch. One basis point was 0.01% of the base execution price. The rates were chosen for a sensitivity comparison and were not calibrated to a particular market. Fees applied from initialization, so different rates could produce different financial histories before the rule switch. Each condition used 50 starting markets on 64×64 grids, with 1,000 baseline steps and 3,000 subsequent steps. We tested all combinations of fraction $f \in \{0.10, 0.15, 0.175, 0.185, 0.20, 0.25\}$, dealer depth $h \in \{0.1, 1, 100\}$, and technical traders active or held. For each fee rate, dealer depth, and technical-trader condition, one no-switch control per starting market retained the original rule and served all six fractions. Each switched run shared its history with this control up to the switch. This comparison used 8,400 simulations. At each simulation step, we recorded the interior Sell fraction, the dealer's quoted price, counts of completed orders and buy orders blocked by insufficient cash, total trader cash, and cumulative dealer fee income. We used these histories to identify Sell-majority formation and calculate maximum drawdown after the rule switch. Fee income remained separate from the dealer's trading reserves. We compared each switched run with its no-switch control at the same fee rate. We also compared each positive fee rate with zero fees under the same rule switch. Both comparisons matched simulations by starting market and held other settings fixed. We used 2,000 bootstrap resamples of the 50 matched starting markets to obtain percentile 95% confidence intervals for these differences. Selected comparisons appear in Supplementary Table S13.

We also compared the main model's unverified signals with a hypothetical information regime in which a trader read Hold instead of a neighbor's Sell intention when that neighbor had no shares. Both observation modes had switched markets and no-switch controls sharing the original baseline history. The comparison measured how the observation rule changed the switch's effect on subsequent raw intentions, completed sales, and drawdown. Its protocol, observation timing, and paired analysis are specified in Appendix A.8.

2.5 Revision timing and interior technical traders

Under the original observation rule, each trader's next intention ignored its own current state and all interior traders updated together, so the market evolved as two separate halves. Every neighbor of a trader occupies the opposite color of a checkerboard, so intentions on the white positions at one step depended only on intentions on the black positions at the previous step, and the reverse. The intentions on white positions at even steps and on black positions at odd steps therefore formed one half of the market, the remaining intentions formed the other half, and neither half entered the other's decisions. With technical traders held, the two halves never influenced each other, whereas with technical traders active they could influence each other only through the technical traders, whose decisions depended on a price set by trades from both halves. Updating every individual at once in a spatial simulation can produce patterns that change under other update schemes [27], so we repeated the main comparisons with a revision probability r. At each step, each interior trader adopted its rule's output with probability r and otherwise kept its current intention, which made its next intention depend on its own state and joined the two halves, and revision probability $r = 1$ recovered the main model.

Along the edge, technical traders adjoined 1.6% of interior traders at grid size $L = 64$, so we also placed technical traders on a fraction ρ of interior positions, in addition to those along the edge (Supplementary Table S16). Interior technical traders followed the same band rule with their own multipliers, traded like other traders, stayed in Hold when technical traders were held, were never given the new rule, and were excluded from the Sell fraction. These comparisons used grid size $L = 64$ and 50 starting markets, with technical traders held at dealer depth $h = 100$ or active at $h = 1$ and 100. We tested revision probabilities $r \in \{1, 0.95, 0.9, 0.75, 0.5\}$ without interior technical traders, and interior fractions ρ of 0.025, 0.05, and 0.10, together with 63 interior positions matching

the number along the edge, with revision probability $r = 1$. At grid sizes $L = 32$ and 128, we repeated revision probability $r = 0.9$ with technical traders held and interior fraction $\rho = 0.05$ with technical traders held or active at dealer depth $h = 100$. Each condition tested 22 fractions f placed around the crossover found in a 20-market pilot. We estimated the midpoint fraction f_{50} by linear interpolation as in the market-size comparison, with 95% intervals from 2,000 bootstrap samples of whole starting markets drawn once for all conditions, so that differences between conditions were paired (Appendix A.7).

2.6 Following shared advice and copying completed trades

To examine crowding, we also ran markets in which some traders followed shared advice or copied another trader's completed trades instead of responding to their neighbors. These markets used the same 64×64 grid, trading, dealer, and technical-trader rules as the rule-switch experiments, without a rule switch or fees: every interior trader used Rule 19280 throughout. After the 1,000-step baseline, we chose 10% or 40% of the interior traders at random as followers and let them follow for the next 3,000 steps. Sixteen other interior traders, also chosen at random, were set aside as traders who could be copied and were counted neither as followers nor as non-followers. Advice was a random sequence of Buy, Hold, and Sell recommendations drawn with probabilities 0.25, 0.50, and 0.25 and unrelated to price or to any trader's holdings. Each recommendation was kept with probability 0.8 at each step and otherwise drawn again. A follower's intention at each step was the current recommendation, which replaced the output of its own rule, and neighboring traders read that intention as they read any other. Followers' orders still required cash or shares and cleared against the dealer like every other order. We compared advice from one source shared by every follower, advice from sixteen sources shared by groups of followers that differed in size by at most one, and separate advice for each follower. When copying, each follower repeated the completed trade that its copied trader made in the previous step, choosing Buy after a completed purchase, Sell after a completed sale, and Hold otherwise. All followers copied one trader from the set-aside group, or groups of followers that differed in size by at most one copied all sixteen.

Each following run was paired with a no-following control that shared its starting market, baseline history, dealer depth, and technical-trader condition, and every outcome compared the same traders in the two markets. To test whether followers' influence on neighboring traders contributed to their financial outcomes, we repeated the advice runs with that influence removed: each non-follower read a neighboring follower's intention from the no-following control, while the followers' advice and orders were unchanged. With technical traders held, this left every non-follower decision identical to the no-following control. In a third comparison, the advice stopped after 1,000 steps, and the followers returned to Rule 19280 without any change to their cash, shares, or the price. After the 3,000 steps of following, every trader with shares sold them to the dealer one share per trader and step, with no purchases, until no further sale completed. We divided each follower's final cash by its cash plus shares at the quoted price when following began, and the follower cash change was the mean of this ratio minus the same mean for the same traders in the no-following control. We also measured the change in the fraction of time that non-followers spent in Sell, over all 3,000 steps and over the final 500, and the change in the maximum drawdown of the quoted price. The main comparison used 50 starting markets not used in any other experiment, with two random assignments of followers and advice per market, dealer depth $h = 1$ and 100, technical traders active or held, and one shared source or separate advice for each follower, giving 5,200 simulations. Before these markets were simulated, we fixed two endpoints for 40% followers of one shared source at dealer depth $h = 100$ with technical traders held: the follower cash change with advice, and its difference from the run in which the followers' influence on neighbors was removed. Each endpoint had to have a positive lower limit on a Bonferroni-adjusted 97.5% interval from 10,000 bootstrap resamples of whole starting markets, computed after averaging the two assignments within each market, and all other intervals were pointwise 95% intervals. Sixteen sources and copying were compared only in an exploratory set of 880 simulations over 10 other starting markets, with 2,000 bootstrap resamples and pointwise intervals.

2.7 Market outcomes, statistical analysis, and selection of examples

We measured intentions and financial outcomes separately because a Sell majority did not necessarily produce a large price decline. The Sell fraction $m(t)$ was the fraction of interior traders in Sell after the intention update at step t, and a run reached a Sell majority if $m(t) \geq 0.5$ at least once

during the observation period after the rule switch. Onset time was the first step after the switch at which this threshold was reached. We estimated majority probability as the fraction of runs reaching the threshold and summarized onset times only for those runs. For dealer depth, market size, and trader-resource comparisons, we also averaged $m(t)$ over the final 500 steps. For the market-size comparison, we denoted this mean by $\bar{m}$ and measured its variation between starting markets using the susceptibility χ_L, the variance of $\bar{m}$ multiplied by the number of interior traders, $(L-2)^2$ [28, 29]. With angle brackets denoting averages over the 50 starting markets at a fixed grid size and parameter combination, we calculated:

$$\chi_L = (L-2)^2 \left[\langle \bar{m}^2 \rangle - \langle \bar{m} \rangle^2\right]. \tag{11}$$

At a fixed grid size, a larger χ_L indicated greater variation in the final mean Sell fraction between starting markets (Eq. 11). For each market size L and condition, we estimated the fractions of traders assigned the new rule at which majority probability reached 25%, 50%, and 75%, denoted by $f_{25}(L)$, $f_{50}(L)$, and $f_{75}(L)$. We used straight lines between probabilities at adjacent tested fractions (linear interpolation), taking the first upward crossing of each probability level and retaining any decreases in the sampled probabilities. The midpoint $f_{50}(L)$ located the response, while a smaller width $f_{75}(L) - f_{25}(L)$ indicated a sharper rise in majority probability. We measured price declines using maximum drawdown, the largest proportional fall from the highest quoted price reached so far within the observation period, including the quoted price immediately before the rule switch. We used price crash to describe a large simulated decline, without a threshold for classifying runs as crashes. We counted orders lacking cash or shares as trader suppression and eligible orders left unfilled as dealer rejection. The reported dealer-rejection fraction was rejected orders divided by eligible orders, using each run's totals after the rule switch, with zero assigned when no orders were eligible. Suppressed and completed order counts covered the same period. We summarized maximum drawdown and dealer-rejection fractions using medians across starting markets under the same conditions. In the fee comparisons, majority effects were differences in probabilities, whereas drawdown effects were medians of the differences within matched pairs. We used 95% Wilson confidence intervals for majority probabilities [30]. Where shown, interquartile bands spanned the 25th to 75th percentiles of outcomes across starting markets under the same conditions. To compare technical traders active and held, we paired runs by starting market and counted the markets that reached a majority under only one of the two conditions. At dealer depths $h = 0.1$, 1, and 100, we added fractions 0.17 and 0.18 from the unchanged-resource runs of the trader-resource comparison, which reproduced the dealer-depth runs exactly where the two comparisons overlapped, and counted each run once. We averaged the paired differences in majority outcome over fixed sets of fractions and obtained 95% intervals from 2,000 bootstrap samples of whole starting markets (Supplementary Table S5).

For the market-size comparison, we added tested fractions to resolve the rise in majority probability, aiming for at least five fractions with estimated majority probabilities strictly between 5% and 95% for each size and condition, allowing up to ten rounds. Each round added the midpoint of every adjacent pair whose lower fraction had a probability below 95% and whose higher fraction had a probability above 5%, using the same 50 starting markets at each added fraction (Supplementary Table S3). For each market size and condition, we drew 2,000 bootstrap samples of 50 starting markets with replacement, keeping the same selection at every tested fraction within a sample. We recalculated f_{25}, f_{50}, f_{75}, the width $f_{75} - f_{25}$, and the maximum susceptibility over tested fractions in each sample, using the 2.5th and 97.5th percentiles for 95% confidence intervals. All requested crossings were bracketed in every bootstrap sample (Supplementary Table S3). To check the sensitivity of these interpolated estimates to the use of straight lines, we fitted a smooth, increasing logistic probability curve and compared its midpoint and width at the same probability levels with the primary estimates. The fitted parameters a_L and b_L controlled the curve's position and steepness, while 0.18 and 0.01 centered and scaled the fraction f for fitting. The fitted Sell-majority probability $p_L(f)$ was given by:

$$p_L(f) = \{1 + \exp\left[-a_L - b_L(f - 0.18)/0.01\right]\}^{-1}, \qquad b_L > 0.$$

We estimated the parameters by maximizing the binomial likelihood of the observed majority counts and refitted the curve in each of the same 2,000 bootstrap samples. Standard errors of its midpoint and width were the standard deviations of their bootstrap estimates (Supplementary Table S11). These intervals and standard errors measured uncertainty from sampling starting markets, conditional on the tested fractions and observation period. They excluded uncertainty from selecting additional fractions and, for interpolated estimates, error from approximating the probability curve by straight lines. A longer observation period would define a different outcome, whose change these uncertainty estimates did not describe.

We selected examples of traders' decisions across the grid from the 3,000-step results, before inspecting images, using proximity to group medians and the lower-numbered starting market to break ties. Selection used 64×64 grids, active technical traders, zero fees, and the baseline resource settings without cash or share interventions. At dealer depth $h = 100$, we selected a run without a majority at fraction $f = 0.15$, both outcomes at $f = 0.175$, and runs with a majority at $f = 0.185$ and 0.20 (Fig. 4). Within each fraction and outcome group, we chose the run nearest the median onset time for runs reaching a majority, or nearest the median Sell fraction averaged over the final 500 steps for runs without a majority. The dealer panels used a single starting market that reached a majority at dealer depths $h = 0.1$, 1, and 100 for fraction $f = 0.20$, chosen to minimize the summed distance from the median onset among majority-reaching runs at each depth. The same starting market also supplied a comparison with technical traders held at dealer depth $h = 100$. We extended the five examples across fractions to 10,000 steps after the rule switch, with snapshots at 0, 1,500, 3,000, 5,000, and 10,000 steps (Fig. 4). The dealer comparisons retained their 3,000-step duration and snapshots at 0, 500, 1,500, and 3,000 steps. Selection therefore represented typical outcomes within each group over the original observation period, and a market without a majority at that endpoint could reach one later. Individual runs illustrated how groups of sellers developed, while probabilities across runs measured how often a majority occurred.

3 Results

We first asked whether a lasting change in how some traders respond to their neighbors could push a market of mixed Buy, Hold, and Sell decisions to a Sell majority. Rule 19286 differs from the baseline Rule 19280 in a single response: a trader with three neighbors in Buy and one in Hold chooses Sell rather than Buy (Fig. 1A,B). We paired each switched market with a no-switch control that kept Rule 19280 and shared its baseline history (Supplementary Table S1). With active technical traders and dealer depth $h = 100$, none of the 50 starting markets reached a Sell majority within 3,000 steps after the switch when the fraction of interior traders given the new rule was $f = 0.10$ or 0.15. All of them did at $f = 0.20$ and 0.25, whereas no control reached a majority over the same period. The median onset fell from 1,070 steps after the switch at fraction $f = 0.20$ to 311.5 steps at $f = 0.25$ (Supplementary Table S2). In the displayed pair at fraction $f = 0.20$, groups of sellers expanded after the rule switch while the control kept its mixture of decisions (Fig. 1C). To test whether a one-time instruction to sell could do the same, we instead set the intentions of the same selected traders to Sell once at the switch and let them keep Rule 19280 for every later update. This one-time instruction produced no majority at fraction $f = 0.15$, 0.20, or 0.25, with technical traders active or held (Supplementary Table S6). Which response changed also mattered. We screened all eighteen rules that differ from Rule 19280 in a single response, including Rule 19286, over 20 starting markets with technical traders active and held. At fraction $f = 0.20$, six rules produced a majority in every market under both conditions. Five of them placed Sell at one of the neighbor sums 1 to 5, the only sums at which Rule 19280 does not already assign Sell, and Rule 19286 was one of these five. The sixth replaced Buy with Hold at neighbor sum 3. The other twelve produced no majority in any market: eight replaced Sell with Buy or Hold, two replaced Hold with Buy at neighbor sums 4 and 5, and two replaced Buy with Hold at neighbor sums 1 and 2 (Fig. 1B; Supplementary Table S7). Thus, the lasting switch to Rule 19286 produced majorities where a one-time instruction to sell, given to the same traders, did not, and outcomes depended on which response changed and how many traders received it.

The rule screen produced majorities with technical traders held as well as active, which suggested that a Sell majority did not require price feedback. To test this across fractions and dealer depths, we repeated the rule switch at 11 fractions between 0.10 and 0.25 with the technical traders either active or held in Hold (Fig. 2). We also varied the dealer depth over $h = 0$, 0.1, 0.5, 1, 2, 5, 10, and 100, because depth sets how far the quoted price moves for a given net trade. In the held condition, trades still completed and, at positive depths, the price still moved, but no decision responded to it. Because no decision in the held condition depended on cash, shares, or price, the onset, final mean Sell fraction, and final interior grid for each starting market and fraction were identical across dealer depths despite different price paths. The held Sell-majority probability was therefore the same at every depth: 0.34 at fraction $f = 0.175$, 0.70 at $f = 0.185$, and 1.00 at $f = 0.20$ (Fig. 2A). Responses to neighbors were thus sufficient to produce a Sell majority. Majority probability rose from zero to one over a narrow range of fractions, which we refer to as the crossover. Within it, the probability was higher with technical traders active than held in 22 of the 24 combinations of dealer

depth and fraction in which the two differed, by at most 0.18, and lower in the other two, by at most 0.04 (Fig. 2A,B). The difference was zero at every tested fraction outside the crossover and at every fraction with $h = 0$, where the price never moved, the technical traders' band kept zero width, and they stayed in Hold. Individual starting markets differed in both directions. At dealer depth $h = 100$ and fraction $f = 0.175$, 13 markets reached a majority only with technical traders active and 5 only when held, and at $f = 0.18$, a fraction not tested in this comparison, the counts were 12 and 14 (Supplementary Table S5). With 95% intervals from resampling whole starting markets in brackets, the active minus held difference at dealer depth $h = 100$, averaged over fractions 0.175, 0.185, 0.19, and 0.195, was 0.09 [0.02, 0.15], and adding fractions 0.17 and 0.18 changed it to 0.04 [−0.02, 0.10] (Supplementary Table S5). Thus, a Sell majority formed without any decision responding to the price, and technical traders along the edge changed which starting markets reached a majority without a consistent change in how often one formed.

We next asked how the spread of selling depended on the number of interacting traders. With technical traders held, we compared markets whose grids had $L = 32$, 48, 64, 96, and 128 rows and columns, adding tested fractions near the rise in majority probability at each size and using the same 50 starting markets at every added fraction (Fig. 3). We repeated the comparison with technical traders active at dealer depths $h = 1$ and 100. In larger grids, majority probability rose more steeply with the fraction f and at smaller fractions. With technical traders held, the fraction at which half the markets reached a majority within 3,000 steps, f_{50}, fell from 0.216 at grid size $L = 32$ to 0.166 at $L = 128$. At every size, both active midpoints were within 0.007 of the held value, usually at or below it, and at $L = 128$ they were within 0.001 (Fig. 3C). These midpoints changed little when a logistic curve replaced interpolation: across all size and technical-trader conditions, the two estimates differed by at most 0.0035, less than a tenth of the shift across grid sizes (Supplementary Tables S3 and S11). The width $f_{75} - f_{25}$, the range of fractions over which the held majority probability rose from 0.25 to 0.75, narrowed about threefold over the same sizes (Fig. 3A). With 95% intervals from resampling whole starting markets in brackets, it was 0.023 [0.018, 0.027] at $L = 32$ and 0.008 [0.004, 0.010] at $L = 128$ (Supplementary Table S3). The susceptibility multiplies the variance of the final 500-step mean Sell fraction between starting markets by the number of interior traders. If the fraction of those 500 steps that each trader spent in Sell were independent of the other traders', the expected susceptibility would be at most 0.25, because each such fraction lies between zero and one. With technical traders held, its largest value over the tested fractions was 21.2 at grid size $L = 32$ and 241.2 at $L = 128$ (Fig. 3B,D), far above that value, so Sell activity was strongly correlated among traders within each market. Observation time also moved the midpoint. At grid size $L = 64$, extending the held markets threefold, from 3,000 to 9,000 steps, lowered it from 0.178 to 0.167. With the same set of tested fractions, increasing the grid size from $L = 64$ to $L = 128$ at 3,000 steps lowered it by about the same amount, 0.011 in both cases, so over these ranges the midpoint depended on observation time about as much as on grid size. Longer observation lowered the midpoint at each size tested this way, $L = 32$, 64, and 128, including when the Sell fraction required to count a run was set to 0.4 or 0.6 instead of 0.5 (Supplementary Table S10). Thus, larger grids sharpened the rise in majority probability and moved it to smaller fractions. The position of the rise also moved with observation time, so the size comparison located a crossover for a finite observation period rather than a fraction that would hold for indefinitely long runs, and these runs did not determine whether such a fraction exists.

In the main model, every interior trader revised its intention at every step from its neighbors' previous intentions and ignored its own, so the market evolved as two checkerboard halves that never influenced each other when technical traders were held. To determine whether the crossover depended on this timing, we let each interior trader adopt its rule's output with revision probability r at each step and otherwise keep its intention, and found that a crossover remained at every tested value but moved to larger fractions as traders revised less often (Supplementary Table S15). With technical traders held, the fraction at which half the markets reached a majority within 3,000 steps was 0.178 at revision probability $r = 1$, 0.298 at $r = 0.9$, and 0.416 at $r = 0.5$. Revising less often joined the two halves: at the tested fractions nearest each crossover, the median difference between the Sell fractions of the two halves late in the observation period was at most 0.014 at every revision probability $r < 1$, compared with 0.254 at $r = 1$. No no-switch control reached a majority at any revision probability, and active technical traders along the edge moved the midpoint by at most 0.004 at either dealer depth. The rise also kept its dependence on market size, because at revision probability $r = 0.9$ the midpoint fell from 0.334 at grid size $L = 32$ to 0.285 at $L = 128$ while the width narrowed about fourfold over the same sizes. In the displayed markets at revision probability $r = 0.9$, majorities again developed

as groups of sellers expanded, without the extended regions of cell-by-cell alternation between Sell and Hold seen at $r = 1$ (Supplementary Fig. S1). Thus, how often traders revised their intentions set where the crossover lay, whereas the crossover itself, its sharpening in larger markets, and the absence of majorities without the rule switch did not depend on updating every trader at once.

Because the midpoint moved with observation time, we extended five example markets with active technical traders and dealer depth $h = 100$ to 10,000 steps after the switch, to see whether markets without a majority at 3,000 steps reached one later and how the others developed (Fig. 4). Before any grid images were inspected, we had selected them from the 3,000-step outcomes as typical of their fraction and outcome. They comprised one market without a majority at fraction $f = 0.15$, one with and one without a majority at $f = 0.175$, and one with a majority at each of $f = 0.185$ and 0.20. The market without a majority at fraction $f = 0.15$ still had none after 10,000 steps, whereas the market without one at $f = 0.175$ reached a majority 4,073 steps after the switch. That market's maximum drawdown rose from 10.17% at 3,000 steps to 26.34% at 10,000 steps (Fig. 4B; Supplementary Table S12). The other three majorities had formed 2,330, 1,836, and 1,028 steps after the switch at fractions $f = 0.175$, 0.185, and 0.20, respectively (Supplementary Table S12). Each displayed majority formed as groups of sellers grew and merged (Fig. 4A,C). At fraction $f = 0.15$, no switched market had reached a majority by 3,000 steps, so any majority there would form late. To measure how often this happened, we extended to 10,000 steps the runs from all 50 starting markets at that fraction, with the same active technical traders and dealer depth, together with their no-switch controls. Two switched markets reached a majority by 10,000 steps, compared with none at 3,000 steps, and no control reached one. The longer runs also showed how the switch affected prices before any majority formed. Every control stayed below a maximum drawdown of 1.15% over 10,000 steps. Among the 50 switched markets at this fraction, the median drawdown was already 5.04% at 3,000 steps, before any of them had a majority, and 5.87% at 10,000 steps. Their largest drawdown was 12.29% at 3,000 steps. At 10,000 steps it was 25.30%, in a different market, one that first reached a majority between 3,000 and 10,000 steps (Supplementary Table S12). Thus, remaining without a majority for 3,000 steps did not establish a long-run state, the rule switch deepened price declines even before any majority formed, and a majority that formed later could raise a market's drawdown well above its value at 3,000 steps.

We next asked how dealer reserves affected the price decline that accompanied a majority, because a dealer with small reserves might be expected to limit a decline by buying fewer of the shares that traders offered. Using the same dealer-depth comparison, we examined the quoted price after the rule switch (Fig. 2). At fraction $f = 0.20$, every starting market reached a majority at every tested depth, with technical traders active or held. The declines nonetheless differed in size: with active technical traders, the median maximum drawdown of switched markets was 99.99% at dealer depth $h = 0.1$, 99.52% at $h = 1$, and 25.15% at $h = 100$ (Fig. 2C). Additional runs at dealer depths $h = 40$ and 400, not among the eight depths above, at the same fraction and with active technical traders, followed the same decrease with depth, with median drawdowns of 48.62% and 7.08% (Supplementary Table S9). Smaller reserves made the quoted price more sensitive to net selling, because the same net sale added a larger proportion to a small share reserve. Shallow positive reserves did not limit these declines by refusing trades. A constant-product dealer never ran out of cash for buying, and its few rejections came from its minimum share reserve or from eligible orders that no longer supported a larger net trade: median dealer-rejection fractions were 226, 34, and 0.6 per million eligible orders at dealer depths $h = 0.1$, 1, and 100 (Fig. 2D). The same price sensitivity also produced large declines without the rule switch. At shallow depth, ordinary order imbalances moved the quoted price up and down by large amounts, so the largest fall from an earlier peak was large even without a lasting trend. No-switch controls at dealer depth $h = 0.1$ had a median drawdown of 77.04%, compared with 0.75% at $h = 100$. The median over starting markets of each switched market's increase in drawdown over its no-switch control rose from 22.96 percentage points at dealer depth $h = 0.1$ to a largest value of 80.31 percentage points at $h = 5$ and fell to 24.39 percentage points at $h = 100$ (Supplementary Table S2). At $h = 5$, the median drawdown was 14.03% in controls and 94.43% in switched markets. The increase was smaller at both ends because at $h = 0.1$ the controls' decline left little room for more, whereas at $h = 100$ the quoted price responded weakly to net selling. Zero dealer depth was a separate case, in which only opposing trader orders matched, the dealer did not trade, and the price stayed fixed, so the drawdown was zero although every market reached a majority. In the displayed starting market 2 at fraction $f = 0.20$, a majority formed at dealer depths $h = 0.1$, 1, and 100 with active technical traders, while the quoted price fell nearly to zero at the two shallower depths and by about a quarter at $h = 100$ (Fig. 5). Thus, at fraction $f = 0.20$, the dealer's reserves set how far the

quoted price fell both with and without the rule switch, and shallow positive reserves did not prevent severe drawdowns.

Trades also depended on the traders' own resources, so we independently multiplied their cash and shares by 0.1, 1, or 10 at the rule switch, at dealer depths $h = 0.1$, 1, and 100 (Fig. 6). We report fraction $f = 0.18$, near the held midpoint of 0.178 for the 64×64 grid used here, and give three neighboring fractions in Supplementary Table S4. Holdings entered the order and clearing calculations but not the interior decision rule, so they could change decisions only through the price response of active technical traders. With technical traders held, every cash and share combination produced the same decision history within a starting market, and the Sell-majority probability was 0.64 at every dealer depth, with a 95% Wilson interval of [0.50, 0.76]. With technical traders active, the probability was 0.58 to 0.60 with unchanged resources, compared with 0.64 held, and ranged from 0.54 to 0.80 across all resource combinations and dealer depths (Fig. 6A). With unchanged resources, some starting markets reached a majority only with technical traders active and others only when held, and the two nearly balanced (Supplementary Table S5). This comparison had no separate no-switch controls, so its drawdowns describe switched markets rather than effects of the switch. Among the 27 active combinations, the five highest majority probabilities all had the share multiplier 0.1, and within each combination of cash and dealer depth that multiplier also gave the smallest drawdown, so a higher majority probability did not bring a larger drawdown (Fig. 6D). With unchanged cash and dealer depth $h = 100$, share multipliers of 0.1, 1, and 10 gave median drawdowns of 3.17%, 20.18%, and 77.55% (Fig. 6B). With unchanged shares, raising the cash multiplier from 1 to 10 raised the median drawdown from 20.18% to 28.48% at dealer depth $h = 100$, but only from 96.46% to 97.10% at $h = 1$, where the decline was already close to complete (Fig. 6B). Dealer rejection did not cushion these declines: median dealer-rejection fractions stayed below 5,000 per million eligible orders in every combination, including those with the largest drawdowns (Fig. 6B,C). Together, the dealer-depth and resource comparisons showed that financial outcomes in switched markets depended on traders' cash and shares and on the dealer's reserves, so a Sell majority did not by itself determine how far the quoted price would fall.

When traders read Hold instead of a selling intention from a neighbor with no shares, mean maximum drawdown fell from 24.36% to 3.23% in the comparison with technical traders held, dealer depth $h = 100$, and switched fraction $f = 0.20$ (Supplementary Fig. S2C and Table S17).

We next asked whether traders who followed a common source of advice, rather than their neighbors, spread their decisions to other traders and gained or lost cash as a result. In these markets no trader received a rule switch, and the advice was a random sequence unrelated to price, so a gain could not come from information about the asset. When 40% of interior traders followed one shared source at dealer depth $h = 100$ with technical traders held, non-followers spent 19.12 percentage points more of their time in Sell than the same traders in the no-following controls, where they spent about a quarter of their time in Sell (Supplementary Table S14A). The followers ended with 26.71 percentage points more cash, relative to their value when following began, than the same traders without following, with a Bonferroni-adjusted 97.5% interval of [25.43, 28.06]. When non-followers instead read each follower's intention from the no-following control, their decisions were unchanged and the followers' gain fell to 2.89 percentage points, so removing the followers' influence on neighboring traders lowered the gain by 23.82 percentage points, with an adjusted interval of [22.30, 25.30]. The gain was a transfer rather than new wealth: after every trader sold its shares, the dealer's cash and the traders' total cash matched the no-following control, so the followers gained what the other traders lost. With shallow dealer reserves the followers lost instead, by 3.12 percentage points at dealer depth $h = 1$ with technical traders held and by 18.08 with them active. Following also deepened price declines: with technical traders held, the maximum drawdown rose over the controls by 27.53 percentage points at dealer depth $h = 100$ and by 55.35 at $h = 1$. When the advice stopped after 1,000 steps, the followers kept a gain of 12.19 percentage points. Non-followers kept selling in some markets but not in others: in 38 of 100 runs they still spent more than 20 percentage points more of the final 500 steps in Sell, whereas in 61 of 100 the increase was below 5 percentage points (Supplementary Table S14B). Separate advice for each follower lowered non-follower selling by 13.00 percentage points and gave the followers a gain of only 4.87. Thus, shared advice spread through neighboring decisions, and followers gained cash at other traders' expense when the dealer had deep reserves mostly because their intentions influenced their neighbors, not because the advice carried information.

In an exploratory comparison over 10 other starting markets, we also varied the number of advice sources and compared advice with copying another trader's completed trades, with 40% followers at dealer depth $h = 100$ and technical traders held (Supplementary Table S14C). Replacing one source with sixteen changed non-follower selling from an increase of 18.90 percentage points to a decrease of 10.92 and lowered the followers' gain from 27.01 to 6.13 percentage points. Copying changed the followers' cash far less than shared advice did, by 0.58 percentage points when every follower copied one trader and 0.81 when groups copied sixteen. Copying sixteen traders lowered non-follower selling by 18.31 percentage points. Copying one trader gave a mean change of 4.07 percentage points with an interval from -6.56 to 16.03, because non-follower selling rose in half of the runs and fell in the other half. Thus, agreement among followers shaped their effect on other traders: one shared source spread selling and brought the largest gains, whereas many sources, separate advice, or copying many traders reduced non-follower selling. These exploratory comparisons were not repeated on new starting markets.

Finally, we asked whether trading costs changed the spread of selling or its financial consequences, repeating the paired rule switch with fees charged on completed orders. With technical traders held, the Sell fraction at every step and the final grid were identical across all four fee rates and three dealer depths for each starting market and fraction, because fees did not enter any decision rule. With technical traders active and fraction $f = 0.20$, the Sell-majority probability across fee rates was 1.00 at dealer depth $h = 0.1$, 0.98–1.00 at $h = 1$, and 1.00 at $h = 100$ (Supplementary Table S13). Nearer the crossover, at fraction $f = 0.175$ and dealer depth $h = 100$, the probability changed from 0.50 without a fee to 0.40 at 25 basis points. The paired difference was -0.10 with a 95% interval of [-0.28, 0.08], so its direction was not resolved. The effect of fees on price declines depended on the condition. At dealer depth $h = 100$ and fraction $f = 0.20$, a 25-basis-point fee lowered the median maximum drawdown from 25.15% to 23.67%, a median paired change of -1.34 percentage points with a 95% interval of [-1.60, -1.16]. At dealer depth $h = 1$ and fraction $f = 0.175$, the same fee raised it from 76.52% to 92.81%, a median paired change of 9.04 percentage points with a 95% interval of [3.46, 16.98]. Over a complete run, the fees accumulated: at 25 basis points, dealer depth $h = 100$, and fraction $f = 0.20$, the median cumulative fee income of the dealer was 36.19% of the traders' initial cash (Supplementary Table S13). Thus, fees reduced the price decline in some conditions and deepened it in others.

4 Discussion

In this study, we explored how a lasting change in the way a fraction of traders formed their intentions could spread selling through a market, and what then determined how far the price fell. Within the observation period, a Sell majority formed once a large enough fraction of traders received the new rule, even though every other trader kept its original rule (Fig. 1). The size of the accompanying price decline depended instead on traders' cash and shares and on the dealer's reserves. The rule switch produced majorities that a one-time instruction to sell, given to the same traders, did not (Supplementary Table S6). The changed response acted whenever a selected trader again had three neighbors in Buy and one in Hold, so the selected traders did not need to remain in Sell for the change to keep influencing the market. Screening every single-response change to Rule 19280 showed that majority formation depended on which response changed, so the conclusions apply to the tested rules rather than to rule switches in general (Supplementary Table S7). Because each switched market shared its history with a no-switch control, the difference between the two measured the effect of the switch in that starting market, and repeating the comparison over starting markets estimated how often a majority formed within the observation period [31, 32]. These conclusions concern assigned rule switches and the spread of selling intentions: traders did not learn the new rule, adopt it for its profitability, or pass the rule itself to other traders.

The spread of selling did not require any decision to respond to the price. With the technical traders kept in Hold, trades still completed and the price still moved, but no decision responded to it, and Sell majorities still formed (Fig. 2). Responding to neighbors' intentions was therefore enough to spread selling, without any trader selling because the price had fallen. In the displayed markets with active technical traders, a majority formed as groups of sellers grew and merged across the grid (Fig. 4). This growth and merger of Sell-rich domains resembles domain growth in interacting systems away from equilibrium [33, 34], and the rise in majority probability with the fraction of traders given the new rule resembles threshold models of how local changes spread through a population [35–38].

Larger grids sharpened that rise, but the fraction at which half the markets reached a majority fell as the observation period lengthened, because some markets without a majority at 3,000 steps reached one later (Fig. 3; Supplementary Table S10). With technical traders held, updating every trader at once also let the two checkerboard halves of the grid evolve separately, and letting traders revise less often joined them and raised the midpoint from 0.178 to 0.416, although a crossover remained at every revision probability tested (Supplementary Table S15). The comparisons therefore located a crossover for finite grids, observation periods, and revision timing rather than a threshold for indefinitely large or long-lived markets [28, 29]. Thus, selling spread through neighboring decisions, and the fraction of traders needed for a majority depended on the grid size, how long the market was observed, and how often traders revised their intentions.

How neighbors interpreted selling intentions after their senders ran out of shares changed the response to the rule switch. In the main model, neighboring states represented public intentions whose financial backing and execution were unverified. A trader could therefore continue to influence others after selling its last share. When we instead let neighbors read Hold from an empty seller, the minority switch's effect on late selling intentions and drawdown was reduced in the primary comparison, while majorities reappeared at higher switched fractions (Supplementary Table S17 and Fig. S2). This counterfactual established sensitivity to one way of using inventory information. It did not establish that real observers could see holdings, that they would respond to disclosure in this way, or that messages without financial backing lacked influence. Public commentary and shared advice motivate distinguishing observable statements from verified transactions, but the model did not represent a media institution or a strategic choice about what to announce. Its findings therefore concerned the spread of communicated intentions under specified observation rules, whose financial effects were measured through completed trades and prices.

How far the quoted price fell, measured by its maximum drawdown, depended on how the dealer's reserves set prices. Dealer models commonly connect trading to intermediary cash and inventory [39–42], but holdings alone do not specify how those resources translate into prices. Here, the constant product of the dealer's cash and share reserves set both the common price at which each step's orders executed and the quoted price after settlement [23]. When the dealer bought shares, its share reserve rose and its cash reserve fell, so continued net selling lowered execution prices. Because the same net sale was a larger proportion of a small share reserve, shallow positive reserves allowed large proportional declines. They still rejected few eligible orders, because a constant-product dealer never ran out of cash for buying (Fig. 2C,D). Zero dealer depth was a separate case: the dealer did not trade, only opposing trader orders matched, and the price stayed fixed, so the absence of a decline reflected the fixed price rather than protection by a small dealer. Markets with shallow reserves also had large drawdowns without the rule switch, so the effect of the switch was the paired increase over each no-switch control. That increase was largest at an intermediate dealer depth rather than at the shallowest (Supplementary Table S2). Thus, a Sell majority did not by itself determine the price decline: its size depended on the execution mechanism and on the resources of traders and the dealer, and shallow positive dealer reserves did not prevent severe declines.

Following shared advice tested a second way in which decisions could spread, in which traders imitated a common source rather than their neighbors. Advice from one source spread selling to non-followers through neighboring decisions, and when the dealer had deep reserves the followers gained cash although the advice was random (Supplementary Table S14). Removing the followers' influence on their neighbors removed most of that gain, and the traders' total cash was unchanged, so the gain came from how other traders responded to the followers and was paid by those traders rather than by the dealer. With shallow reserves the same advice left the followers with less cash, and in the exploratory comparison many sources, separate advice, or copying many traders reduced rather than increased selling by non-followers. Herding models have linked imitation to large aggregate fluctuations [8, 43], and here imitation of a shared source also moved cash from the other traders to the followers. Thus, crowding around a single source changed both how selling spread and who gained or lost, whereas the followers' average gain became a loss when the dealer's reserves were shallow.

Compared with established agent-based markets, the model differed in how traders interacted and how their rules changed: we assigned a persistent rule switch instead of modeling learning or strategic choice. In the minority game, traders choose among strategies scored against shared past outcomes and interact through the market outcome [44, 45]. In the Santa Fe artificial stock market, traders learn forecasting rules from a shared price and dividend [5]. Related studies of the Santa Fe market

examined incentives to adopt technical rules [46] and choose how often to revise forecasts [47]. Models with chartists and fundamentalists let traders move between the two groups at rates set by recent profits or by others' opinions [7, 9, 48]. Lattice and cellular-automaton markets encoded local interactions through stochastic imitation or spin-like responses, sometimes alongside market-wide signals [10, 11, 17–20, 49]. Wolfram's one-dimensional market and a later model built on Rule 110 used deterministic rules and set the price from the numbers of buyers and sellers [14, 15]. Here the intervention changed one response for a chosen fraction of traders, and the paired control retained the baseline rule. In the main model, only technical traders along the edge responded to price, allowing the rule switch to be tested with price feedback active or removed. Orders needed cash or shares and were completed by other traders or by a dealer with limited reserves, so intentions could spread without a matching increase in completed sales, and a Sell majority and a price decline were separate outcomes. These choices gave up the learning and strategic behavior of established models in exchange for a finite rule set, in which every single-response change could be screened, and an intervention that could be repeated exactly. The questions answered were correspondingly narrower: not whether technical trading emerges or whether the market becomes efficient, but what fraction of traders given a new rule produced a Sell majority within a finite observation period, whether its formation required feedback from price, and how resources and execution determined the resulting price decline. The following experiments asked, in the same narrow sense, whether followers of shared advice gained through their influence on other traders.

Several modeling choices limit how far these conclusions extend. Traders followed assigned rules rather than choosing strategies to maximize returns, and they used one baseline lookup table on a regular grid, in which different neighbor configurations with the same sum produced the same response. The inventory comparison tested one response to observing whether a sender held shares, while other degrees of disclosure, responses to that information, and limits on Buy signals were untested. The only traders whose decisions depended directly on the price were technical traders following one band rule, who bought when the price fell below its recent range and sold when it rose above it. Making them active rather than held moved the fraction needed for a majority by less than 0.01 when they stood only along the edge, but by 0.149 at dealer depth $h = 100$ when they also occupied 5% of interior positions, and this difference stayed between 0.14 and 0.15 at grid sizes from $L = 32$ to 128 (Supplementary Tables S3 and S16). We did not test traders who sell because the price has fallen, such as trend-following or stop-loss rules, so the model cannot show whether falling prices would accelerate the spread of selling. Unit orders cleared at a common execution price against a constant-product dealer, a specific reserve mechanism rather than a calibrated representation of an equity exchange, whereas other models link order flow to prices through dealer inference or order-book dynamics [50–52]. Varying dealer depth changed the dealer's funding and its initial price sensitivity together, so the depth comparisons could not separate the two effects (Supplementary Table S9). The dealer collected prescribed fees in the fee comparisons but did not set its charges or its participation to maximize profit, and we did not model incentives to provide or withdraw liquidity. The model had no order book, dividends, borrowing, or strategic risk management by the dealer, and its steps and price response were not fitted to market data, so its simulated crashes did not predict the timing or magnitude of actual market crashes. The advice given to followers was random, and copying and the number of advice sources were compared only in an exploratory set of starting markets, so we did not test advice that carried information about the asset or copying of traders chosen for their past performance. Other decision rules, updates in random order, alternative interaction networks, and different execution mechanisms would test how the connection between local decisions and price declines depends on these assumptions.

5 Conclusion

A lasting change in how a fraction of traders responded to their neighbors could move a market from mixed decisions to a Sell majority, even when no trader's decision responded to the price. The fraction needed fell in larger markets and when the market was observed for longer, and rose when traders revised their decisions less often, so it marked a crossover for finite markets, observation periods, and revision timing rather than a fixed threshold. Under the original observation rule, neighbors read publicly communicated intentions without verification of holdings or trades. Reading Hold from a seller with no shares reduced the minority switch's effect on late selling intentions and drawdown in the primary comparison, while majorities reappeared at higher switched fractions. A Sell majority did not by itself determine how far the price fell. Trader cash and shares limited which orders could

be completed, and the dealer's cash and share reserves set the prices at which the remaining orders cleared. Shallow positive dealer reserves allowed severe declines with or without the rule switch, whereas zero dealer depth kept the price fixed because the dealer did not trade. Small dealer reserves therefore did not protect against a crash, although giving traders fewer shares reduced the decline. Traders who followed one shared source of random advice spread selling through their neighbors and, when the dealer's reserves were deep, gained cash that the other traders lost. The financial consequence of widespread selling depended on how traders interpreted public intentions and how the resulting orders were executed and priced.

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Author contributions. J.D.V. designed the model, wrote the code, performed the experiments, and wrote the paper.

Competing interests. The author declares no competing interests.

Code and data availability. The simulation code, analysis scripts, figure and table generators, and the campaign outputs used in this manuscript are available under the MIT license at `https://github.com/varnerlab/Wolfram-markets-abm-paper`. The numbers reported here come from the archived studies in the repository's `output/rebuilt` directory, whose manifests record source hashes, parameters, and package versions.

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Figures

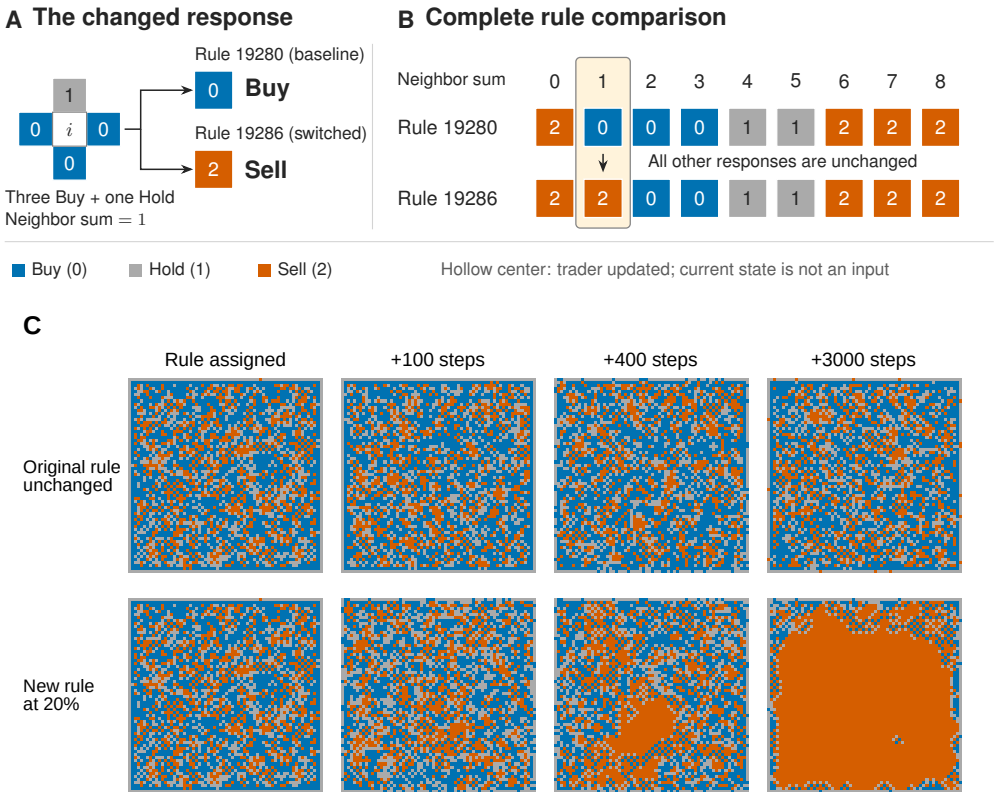


Figure 1: The changed response and paired market decisions. **(A)** The same four neighboring intentions produce different responses at the trader marked i: three Buys and one Hold give neighbor sum 1, producing Buy under Rule 19280 and Sell under Rule 19286. The hollow center's current intention is not an input. Every arrangement with the same neighbor sum produces the same response. **(B)** Complete lookup tables, with the baseline and switched rules aligned by neighbor sum. The highlighted column is the only response changed by the switch. Blue, gray, and orange denote Buy (0), Hold (1), and Sell (2); arrows indicate rule evaluation or a changed response, not completed trades. **(C)** Paired 64×64 grids at the rule switch and after 100, 400, and 3,000 steps. The upper row retains the baseline rule; the lower assigns the new rule to 20% of interior traders. Both markets share a 1,000-step baseline, dealer depth $h = 100$, and active technical traders. Starting market 14 was selected from 50 by proximity to the median paired difference in peak Sell fraction. Ensemble outcomes and conditional onset times appear in Supplementary Table S2.

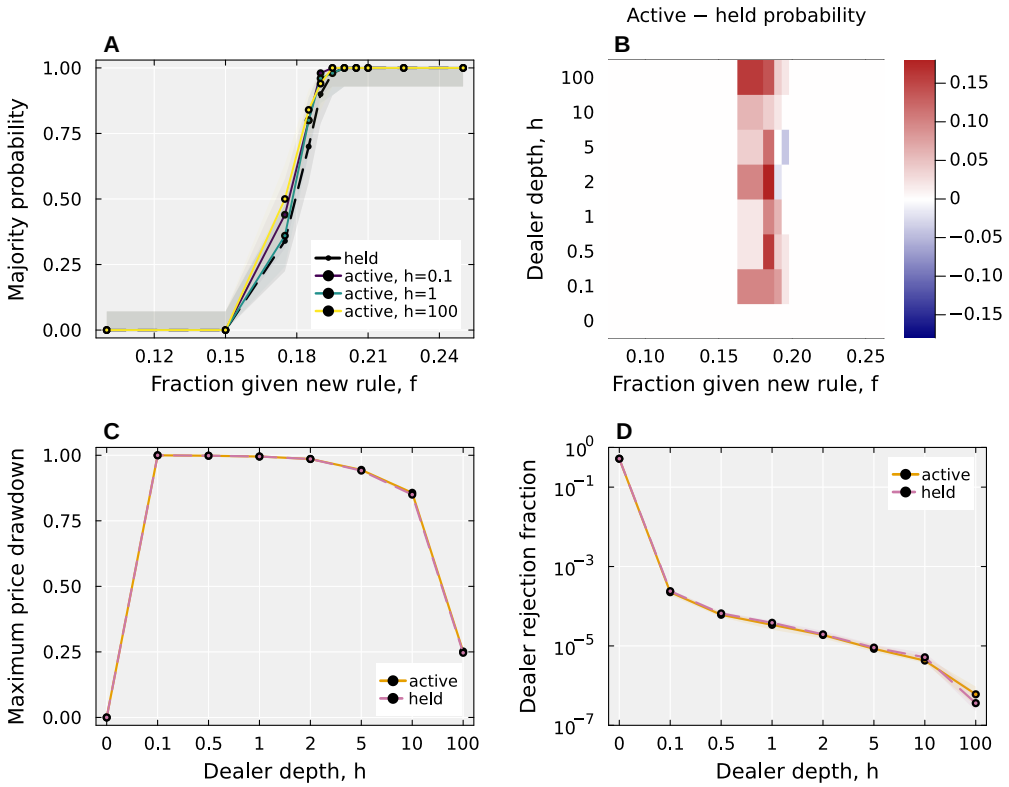


Figure 2: Rule switches across dealer depths over 3,000 post-switch steps. **(A)** Probability of an interior Sell majority with technical traders held, or active at dealer depths $h = 0.1$, 1, and 100. Bands are 95% Wilson intervals over 50 starting markets. **(B)** Active-minus-held majority probability at each depth and fraction. **(C)** Median maximum drawdown at fraction $f = 0.20$, with interquartile bands. **(D)** Median fraction of post-switch eligible orders rejected at fraction $f = 0.20$, on a logarithmic axis with interquartile bands. Dealer depth h is the dealer's initial shares per trader. Because the dealer's cash and share reserves keep a constant product, h also sets how far the quoted price moves for a given net trade. At zero dealer depth only opposing trader orders match and price stays fixed. Panels C and D describe switched markets. No-switch control drawdowns and paired differences appear in Supplementary Table S2.

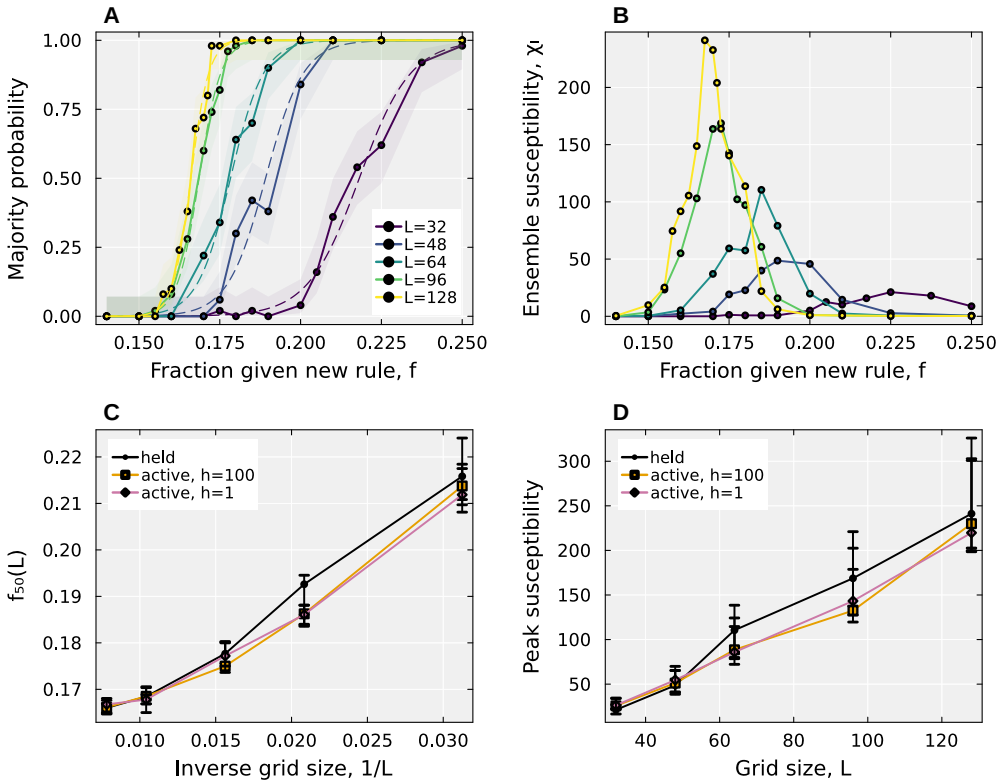


Figure 3: Market size and the formation of a Sell majority over 50 starting markets and 3,000 post-switch steps. **(A)** Majority probability with technical traders held, with 95% Wilson bands. **(B)** Susceptibility with technical traders held: the variance across markets of the final 500-step mean Sell fraction, multiplied by the number of interior traders. **(C)** Interpolated fraction $f_{50}(L)$ at which half the markets reached a majority, against inverse grid size, with technical traders held at dealer depth $h = 100$ and active at $h = 1$ and $h = 100$. **(D)** Sampled maximum susceptibility versus grid size. Error bars in C and D are 95% whole-market bootstrap intervals. Solid lines connect tested values; dashed curves in A are logistic fits. No law for indefinitely large grids was fitted.

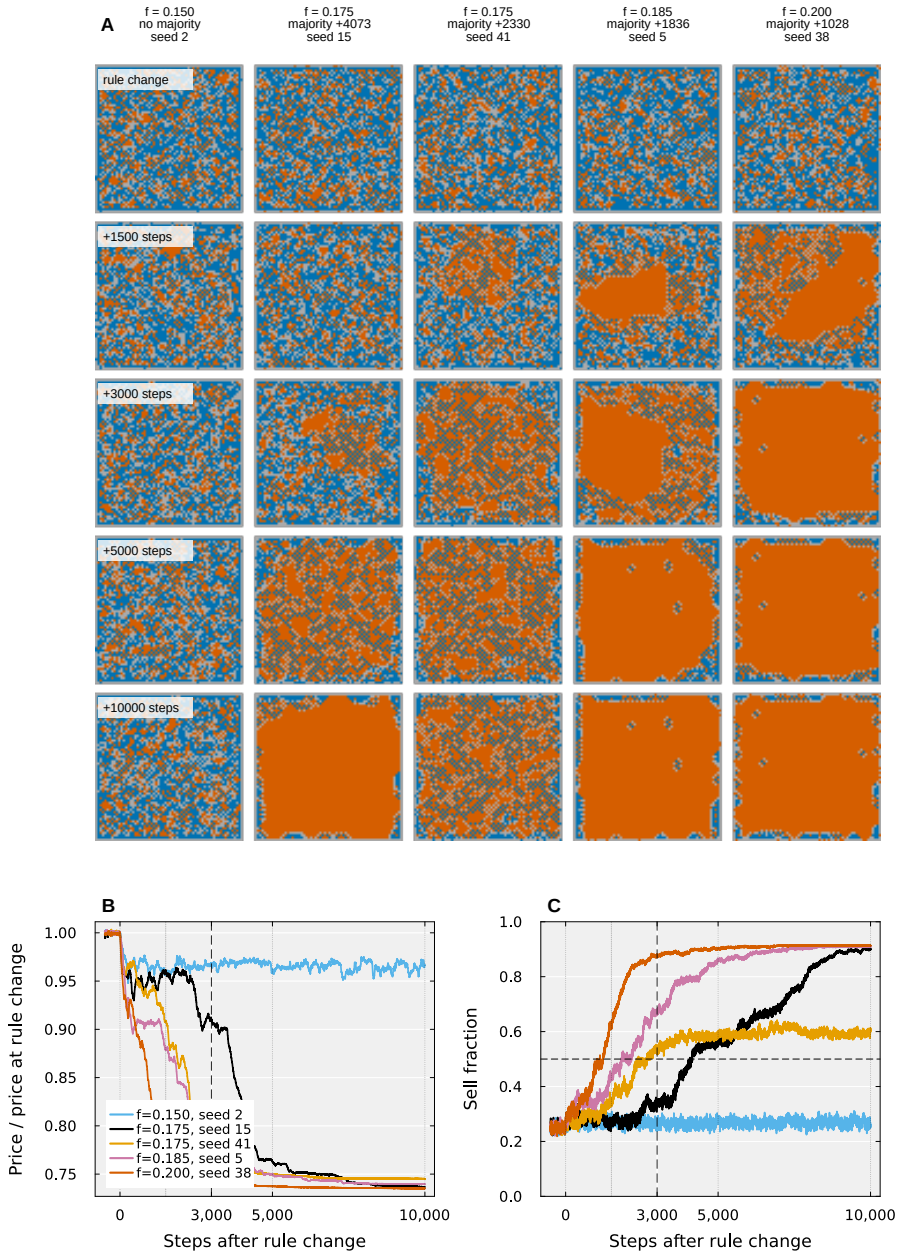


Figure 4: Five example markets extended to 10,000 steps with active technical traders and dealer depth $h = 100$. Blue, gray, and orange denote Buy, Hold, and Sell. **(A)** Grid decisions at the rule switch and after 1,500, 3,000, 5,000, and 10,000 steps. The examples were selected from the 3,000-step summaries before any images were inspected. At that endpoint, the first two markets had not reached a majority and the other three had. Nonmajority examples were nearest their group's median final Sell fraction; majority examples were nearest their group's median onset, with the starting-market number breaking ties. Starting markets were 2, 15, 41, 5, and 38, from left to right. The longer outcomes did not affect selection. Because each trader's update ignores its own intention, the two checkerboard halves of the grid can hold different intentions, which appears as regions where Sell and Hold alternate from cell to cell; no such extended regions appear in the example markets with revision probability $r = 0.9$ (Supplementary Fig. S1). **(B)** Quoted price divided by its value at the rule switch. **(C)** Interior Sell fraction, with one-half marked by a horizontal line. The dark vertical dashed line marks the original 3,000-step endpoint; other guides mark the remaining snapshots. Onsets and financial summaries appear in Supplementary Table S12.

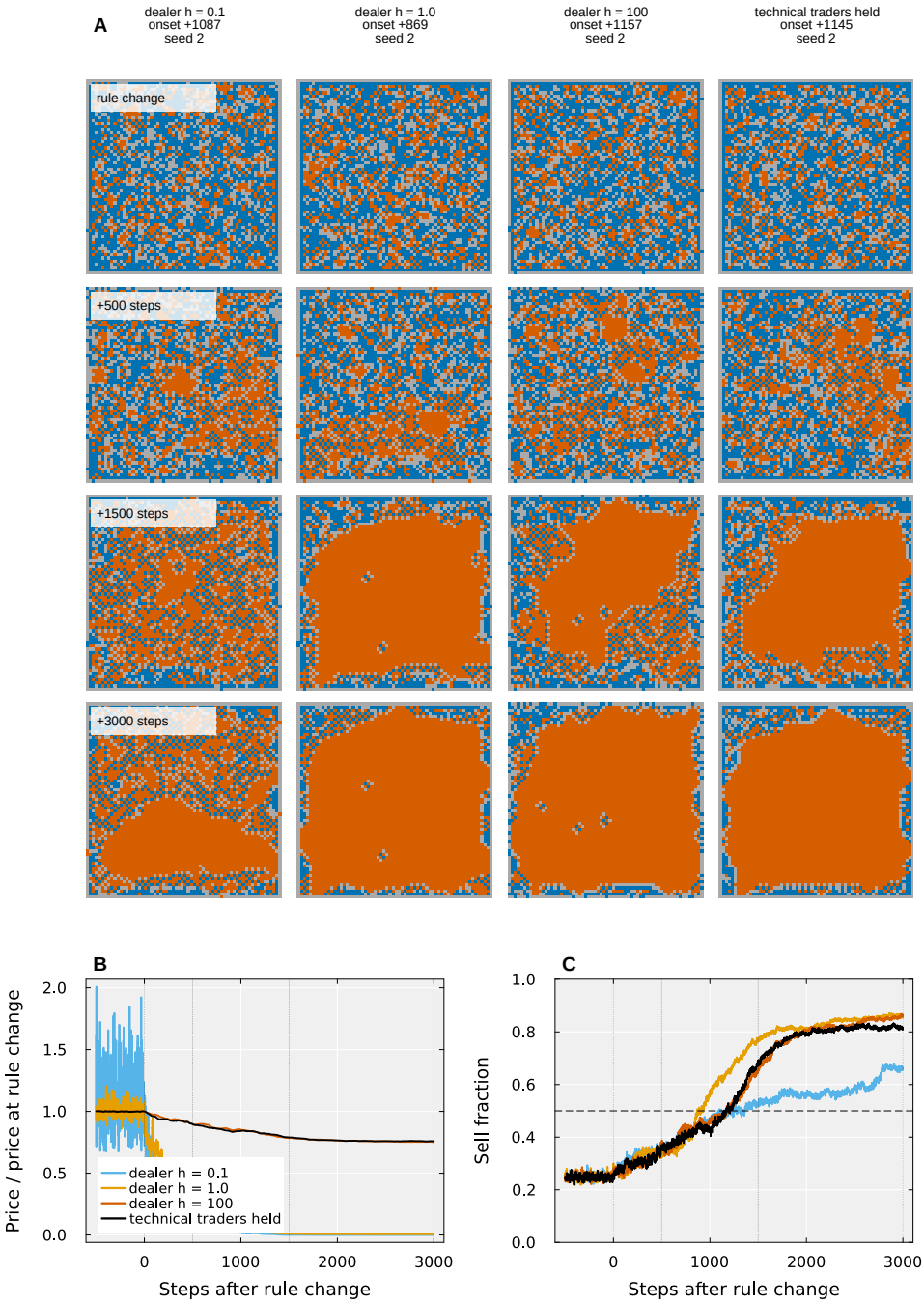


Figure 5: The same starting market at different dealer depths with fraction $f = 0.20$. **(A)** Grid decisions at the rule switch and after 500, 1,500, and 3,000 steps. The first three columns have active technical traders at dealer depth $h = 0.1$, 1, and 100; the last holds the technical traders at $h = 100$. Starting market 2 was selected among markets reaching a majority at all three dealer depths with active technical traders by minimizing the summed distance from their depth-specific median onsets, with the starting-market number breaking ties. **(B)** Quoted price divided by its value at the rule switch. **(C)** Interior Sell fraction. With technical traders held, no decision responds to the quoted price, while interior traders still trade and the quoted price still updates. The four runs share the same starting market; ensemble probabilities and drawdowns appear in Fig. 2.

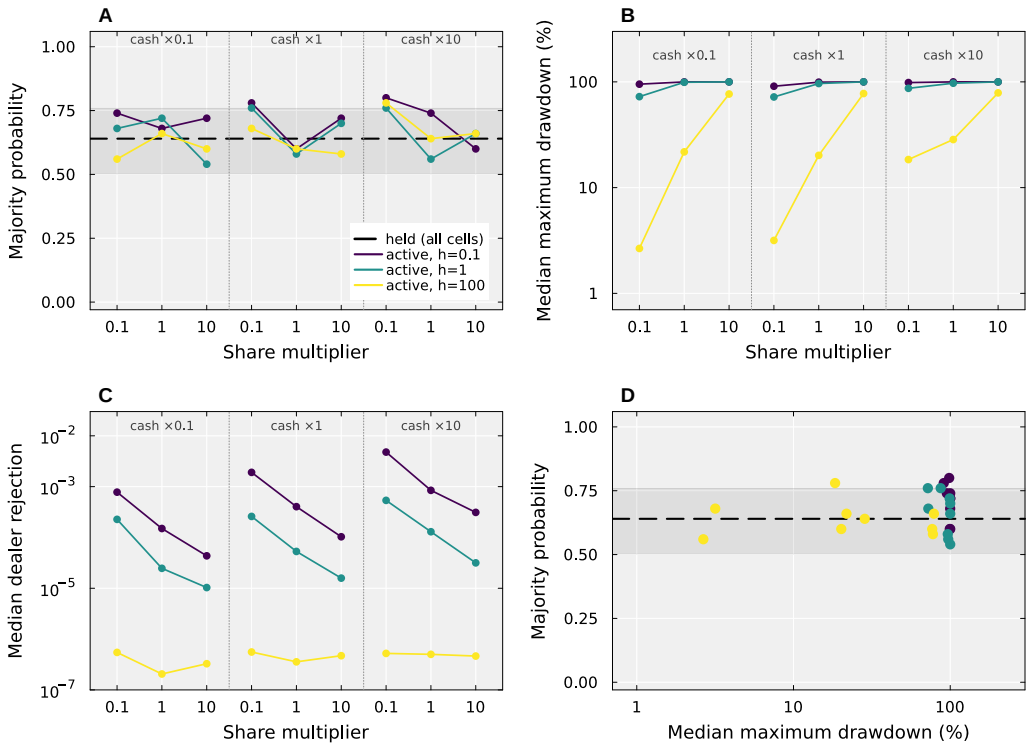


Figure 6: Changing trader cash and shares at the rule switch, with active technical traders, fraction $f = 0.18$, and 50 starting markets over 3,000 post-switch steps. In A–C, the nine resource combinations are grouped by cash multiplier, with share multiplier on the axis and a series for each dealer depth. **(A)** Majority probability. The dashed line and band give the held-condition probability and its 95% Wilson interval, identical across resource combinations and dealer depths. **(B)** Median maximum drawdown of switched markets on a logarithmic axis; this comparison had no no-switch controls. **(C)** Median fraction of post-switch eligible orders rejected, on a logarithmic axis. **(D)** Majority probability against median drawdown for all 27 active combinations, with the held reference repeated. The drawdown axis follows B. Probability axes cover the full possible range; all displayed drawdowns and rejection medians are strictly positive.

A Supplementary methods and validation

A.1 One complete simulation step

The main simulation separated a desired decision from its execution. Orders used the decisions stored before the interior update, while technical traders used the new quoted price before the next interior decisions were calculated. A resource shortage could therefore stop an order without changing the state seen by neighboring traders. The uniform execution price depended on the net quantity supported by the entire batch (Eqs. 3 and 4), and differed from the quoted price left after settlement (Eq. 8). Executed counts specified the net transfer and its normalized imbalance (Eqs. 5 and 6), which changed the dealer reserves (Eq. 7). The new quoted price then updated the reference band (Eq. 10). Algorithms S1 and S2 specify this ordering for the main persistent-rule experiments with adaptive bands and no cost-basis restriction, including the transaction-fee extension. The pulse, alternative-rule, and frozen-band controls modify the corresponding updates as described in Appendix A.5. The counterfactual observation rule in Appendix A.8 additionally replaced a zero-share sender's Sell signal with Hold before the interior update, while preserving its stored intention and the settlement order.

Algorithm S1 One market step, $t-1 \to t$, in simulation order. Stored decisions generated orders. Clearing determined a common execution price, and the settled dealer reserves set the new quoted price.

Require: grid X_{t-1}; technical-trader decisions; trader books; dealer (W_{t-1}, Q_{t-1}); fee account G_{t-1} and rate τ; quoted price and band state $(P_{t-1}, \mu_{t-1}, \sigma_{t-1})$; rule field R_i; parameters; seed
Ensure: grid X_t; technical-trader decisions; settled books and dealer; quoted price and band state (P_t, μ_t, σ_t); order counts

if t is the scheduled rule-switch step **then**
  $R_i \leftarrow 19286$ for the preselected interior traders {preserve current decisions and financial state}
end if
if t is a scheduled resource intervention **then**
  for all traders i **do**
    $w_i \leftarrow g_w w_i$; $q_i \leftarrow \mathrm{round}(g_q q_i)$
  end for
end if
call Algorithm S2 using the stored decisions
receive settled books, dealer and fee account, order counts, base price F_t, and quoted price P_t
$\mu_t \leftarrow \mu_{t-1} + \alpha(P_t - \mu_{t-1})$
$\sigma_t \leftarrow \sigma_{t-1} + \alpha(|P_t - \mu_t| - \sigma_{t-1})$
if technical traders are active **then**
  for all technical traders i **do**
    $e_i(t) \leftarrow 0$ if $P_t < \mu_t - k_i\sigma_t$; 2 if $P_t > \mu_t + k_i\sigma_t$; 1 otherwise
  end for
else
  $e_i(t) \leftarrow 1$ for every technical trader i
end if
write $e_i(t)$ to the technical traders at selected edge positions; leave other edge positions in Hold
for all interior traders i, simultaneously **do**
  $c_i \leftarrow \sum_{j \in \partial i} s_j$ using old interior and new technical-trader decisions
  $s_i(t) \leftarrow \mathcal{R}_{R_i}(c_i)$
end for

Algorithm S2 Reserve clearing at a common base price without a cost-basis restriction. The binary search found the greatest supported integer net purchase in the direction of initial excess demand. The baseline uses $\tau = 0$.

Require: desired unit buys and sells from stored decisions; trader books $(w_i, q_i, \bar{c}_i)$; dealer (W, Q); fee account G; fee rate τ; previous quoted price; seed and step
Ensure: settled books and dealer; order counts; execution price and next quoted price
if dealer depth $h = 0$ **then**
  use the previous quoted price F; match buys with $w_i \geq (1+\tau)F$ and sells with $q_i \geq 1$
  apply priorities; use the buyer and seller cash, share, and cost-basis updates shown below
  update G as below; keep trading reserves at zero and the quoted price fixed
  return settled books, fee account, and order counts
end if
define $F(x) = W/(Q - x)$ for integer x satisfying $Q - x \geq 1$
define $B(x)$ as desired buyers with $w_i \geq (1+\tau)F(x)$ and $S(x)$ as desired sellers with $q_i \geq 1$
$d \leftarrow \operatorname{sign}[B(0) - S(0)]$; $\ell \leftarrow 0$
$u \leftarrow \min(B(0) - S(0), Q - 1)$ if $d > 0$, otherwise $S(0) - B(0)$
while $\ell < u$ **do**
  $k \leftarrow \lfloor (\ell + u + 1)/2 \rfloor$
  if $d[B(dk) - S(dk)] \geq k$ **then**
    $\ell \leftarrow k$
  else
    $u \leftarrow k - 1$
  end if
end while
$x \leftarrow d\ell$; $F_t \leftarrow F(x)$
collect eligible sets $\mathcal{B}$ and $\mathcal{S}$ at F_t
$n^B \leftarrow \min(|\mathcal{B}|, |\mathcal{S}| + x)$; $n^S \leftarrow n^B - x$
suppression $\leftarrow$ desired counts minus eligible counts
rejection $\leftarrow$ eligible counts minus executed counts
rank any rationed side by $z(\text{seed}, t, i, \text{side})$, breaking ties by identity
for all selected buyers i **do**
  $\bar{c}_i \leftarrow [\bar{c}_i q_i + (1+\tau)F_t]/(q_i + 1)$; $q_i \leftarrow q_i + 1$; $w_i \leftarrow w_i - (1+\tau)F_t$
end for
for all selected sellers i **do**
  $q_i \leftarrow q_i - 1$; $w_i \leftarrow w_i + (1-\tau)F_t$
end for
$Q' \leftarrow Q - x$; $W' \leftarrow WQ/Q'$; $P_t \leftarrow W'/Q'$
$G' \leftarrow G + \tau F_t(n^B + n^S)$
return settled books, dealer, fee account, counts, F_t, and P_t

A.2 Rationing and paired reproducibility

Each eligible order on a rationed side received a 64-bit score from a SplitMix64 finalizer applied to the simulation seed, step, trader identity, and order side. Sorting by score, then identity, made rationing reproducible while allowing priority to change over time. Paired markets used the same score for the same trader and step even after their eligible sets diverged. Scoring consumed no draws from the generator used for initialization. For paired rule switches, comparisons checked the pre-switch grid and Sell history, technical-trader coordinates and multipliers, and price history through the first post-switch quoted price. A pure rule switch preserved that first quoted price because the stored decisions still generated the same orders. A one-time Sell pulse and a resource intervention could change the first batch and therefore were required to agree only before the intervention. For held technical traders, resource comparisons also required identical Sell histories and final grids across all cash and share multipliers.

A.3 Conservation and limiting cases

Let total trader cash and shares be the sums over all traders. Outside scheduled resource interventions, their combination with dealer trading reserves and the fee account satisfied the conservation identities:

$$\sum_i w_i(t) + W_t + G_t = \sum_i w_i(0) + W_0 + G_0,$$
$$\sum_i q_i(t) + Q_t = \sum_i q_i(0) + Q_0.$$

The production wrapper checked these identities separately before and after an intentional resource change. Share conservation was exact. The allowed absolute cash discrepancy was the larger of 10^{-5} and 10^{-11} times total cash at the beginning of the checked interval. For positive dealer depth, checks required a finite positive quoted price, nonnegative trader resources, preservation of the reserve product to relative 10^{-9}, and agreement between the quoted price and reserve ratio. The implementation evaluated new dealer cash as WQ/Q' to limit accumulated rounding error; the equivalent transfer was $F_t x_t$. These checks ran after every newly completed simulation, before its execution record was written. The original experiments used $G_t = 0$. The fee extension included cumulative dealer revenue in cash conservation and checked that revenue against fees on completed orders. Fees were excluded from the reserve-product and quoted-price checks.

Zero dealer depth permitted only matched trader orders and kept price fixed. Technical traders then remained in Hold under either technical-trader condition, providing an exact comparison between active and held interior decisions. At positive dealer depth, tests checked settlement at the selected price, affordability after clearing, discrete rationing, and isolated round trips that restored dealer inventory and trading cash. With a positive fee, the isolated round trip also transferred cash from the trader to the dealer's fee account. The fee tests checked zero-fee agreement with the frozen original engine, affordability including fees, conservation, balanced orders, and the fixed-price zero-reserve case. They also covered a delayed rule switch, preservation of a common prehistory, and the held-decision invariance across financial interventions. An independent implementation of the held decision update served as a test oracle. Every production condition, including held decisions, was nevertheless rerun with the full financial simulator. Finite dealer depths were distinct model conditions: no finite-depth path was assumed to equal an unlimited dealer.

A.4 Simulation design and reproducibility

The clean rebuild began with an empty data and figure tree and a frozen copy of the corrected simulator, drivers, parameter manifest, and analysis code. No earlier simulation output or checkpoint was imported. Completed simulations wrote an execution ledger, while phase manifests and source hashes constrained any resumption to outputs created within this rebuild. All retained experiments were rerun, including held conditions whose decisions could be predicted without calculating finances. Seeds were retained to reproduce the planned starting markets; this was a complete recomputation under the corrected model, not an independent sample of starting markets. The corrected model replaced settlement at a standing quote followed by a separate exponential price update. That earlier mechanism admitted profitable isolated round trips, and its financial outputs did not enter this manuscript. Reserve clearing changed both settlement and price formation, so differences from the earlier model cannot be attributed to one parameter alone.

The four main campaigns comprised 30,550 simulations over 50 starting markets per condition and 3,000 post-switch steps (Supplementary Table S1). The selected paired-switch campaign used grid size $L = 64$, dealer depth $h = 100$, and active technical traders at fractions 0.10, 0.15, 0.20, and 0.25. Dealer comparisons used fractions 0.10, 0.15, 0.175, 0.185, 0.190, 0.195, 0.200, 0.205, 0.210, 0.225, and 0.250 at dealer depths $h = 0, 0.1, 0.5, 1, 2, 5, 10, 100$ with technical traders active and held. A newly simulated no-switch path was shared across fractions within each dealer depth, technical-trader condition, and starting market, avoiding duplicates within the same campaign. The size campaign used grid sizes $L = 32, 48, 64, 96, 128$ at fractions 0.140, 0.150, 0.160, 0.170, 0.175, 0.180, 0.185, 0.190, 0.200, 0.210, 0.225, and 0.250 before refinement. Its conditions were technical traders held at dealer depth $h = 100$ and technical traders active at $h = 1$ and $h = 100$. The cash and share campaign combined independent multipliers 0.1, 1, and 10 with fractions 0.170, 0.175, 0.180, and 0.185, dealer depths 0.1, 1, and 100, and both technical-trader conditions. Dealer resources increased with the trader count; no size series with fixed absolute dealer resources was included.

The supplementary controls retained these starting-market conventions while varying the intervention, dealer depth, band adaptation, or observation period. Their fixed design preceded this clean rebuild; adaptive refinement and example selection followed the numerical rules specified above. We used Julia 1.12.7 with the repository environment manifest, twelve simulation threads, and one garbage-collection thread. A seeded Xoshiro generator supplied setup draws [53]. Checkpointed drivers wrote atomic records and reconstructed a deterministic row order. A recorded scheduling-only revision assigned supplementary jobs to available workers individually. Validation against serial execution and the original schedule reproduced complete market histories exactly; the simulation equations, parameters, and analysis were unchanged. Python performed the bootstrap analysis and generated tables from the new summaries. The accompanying repository snapshot contains source hashes, parameter manifests, execution records, package versions, and derived outputs. The project repository is `https://github.com/varnerlab/Wolfram-markets-abm-paper`. The repository retains earlier implementations and outputs as a record of model development, outside the rebuild's permitted data inputs.

The example trajectories used newly computed 3,000-step summaries to select typical markets before inspecting their images. We reran the five selected conditions and all 50 starting markets at fraction $f = 0.150$ for 10,000 post-switch steps, each with a matched no-switch path (Fig. 4; Supplementary Table S12). Selection was not repeated using the longer histories. Comparisons at the original endpoint checked reproduction of the new short-run summaries; additional figure reruns checked full histories and final grids for the selected cases. The longer runs extended observations of the same starting markets and did not supply independent replications. Majority onset began with the first updated state after the rule switch, and all reported execution counts and dealer-rejection fractions used post-switch steps. Keeping these measurement windows common avoided mixing baseline trading into one campaign's financial summaries but not another's.

The completed rebuild was checked against its fixed parameter manifest and its execution ledger. All fixed conditions were present, and the new size and horizon campaigns with technical traders held agreed for 1,800 overlapping starting-market and fraction combinations. Across the main dealer, size, and resource campaign summaries, the largest absolute cash discrepancy was 3.40×10^{-6}, and share discrepancies were zero. These numerical checks accompanied the per-simulation reserve and conservation checks described above.

A.5 Additional controls and sensitivity analyses

The supplementary controls used the same initialization and baseline as the main comparisons, with grid size $L = 64$, dealer depth $h = 100$, and 3,000 post-intervention steps unless otherwise specified. Every switched run was compared with a no-switch control. Held conditions calculated full financial histories as well as decisions. A Sell pulse wrote Sell at the selected interior traders immediately before the first post-baseline orders, while retaining the baseline rule for later updates. Traders already in Sell were unchanged. A persistent intervention instead changed the selected rule without overwriting decisions, so its first price remained identical to the no-switch price. These paired interventions compared the same fractions and starting markets under both technical-trader conditions (Supplementary Table S6). The rule screen tested all eighteen ways of changing one response in the baseline lookup table, using twenty starting markets per condition and three fractions. Each screen entry was descriptive rather than a separate confirmatory hypothesis test (Supplementary Table S7).

The band control froze the recent-departure scale at its value immediately before the rule switch, while each technical trader retained its multiplier and the reference price continued to update. Adaptive and frozen runs shared their complete pre-switch histories. The frozen widths could therefore differ across dealer depths and starting markets. At fraction $f = 0.175$, the estimated majority-probability difference between dealer depths $h = 0.1$ and $h = 100$ was -0.06 with adaptive widths and -0.26 with frozen widths. Subtracting those two differences gave 0.20, with a 95% whole-market bootstrap interval of [-0.08, 0.50] (Supplementary Table S8). This comparison changed width adaptation while retaining the response to departures from a moving reference price. It therefore tested one part of the price feedback rather than removing technical trading.

Additional comparisons used dealer depths $h = 40, 100, 400$ at fraction $f = 0.20$ with active technical traders (Supplementary Table S9). For the constant-product dealer, differentiating log price with respect to normalized net trader purchases at initialization gives $2N_{\mathrm{tr}}/H$, approximately $2/h$. The chosen depths therefore spanned initial slopes 0.05, 0.02, and 0.005. No independent

exponential impact coefficient was varied: changing depth also changed dealer funding. These comparisons replaced an impact-coefficient sweep belonging to the superseded standing-quote model. They measured sensitivity within the corrected mechanism and did not hold available reserves fixed while changing price sensitivity. Neither the quoted price nor the length of a step was calibrated to an observed market.

The horizon campaign used fifty starting markets at fractions 0.140, 0.145, through 0.250, extending to 0.300 for grid size $L = 32$ to bracket short-horizon responses. Held technical traders prevented financial feedback throughout. Endpoint comparisons used the original 3,000 steps and size-dependent observation periods $3000(L/64)^z$ for $z = 1, 2$, alongside longer observations. Majority criteria of 0.4, 0.5, and 0.6 were extracted from each history. Interpolation and bootstrap intervals were calculated separately for every size, horizon, and criterion (Supplementary Table S10). The no-switch extension followed ten starting markets per technical-trader condition for 100,000 additional steps. The observed outcomes were active technical traders: 0/10 majorities, 95% Wilson interval [0.000, 0.278]; held technical traders: 0/10 majorities, 95% Wilson interval [0.000, 0.278]. These extensions reused starting markets and did not constitute independent replications. The observed frequencies of Sell majorities described the stated interval without establishing behavior at indefinitely long times.

We also characterized the shared pre-switch histories at dealer depth $h = 100$ with active technical traders. Across the fifty starting markets, mean interior Sell fractions were 0.2494 over steps 501–750 and 0.2504 over steps 751–1,000. In the latter interval, the median within-market log-return standard deviation was 0.000323, and median within-market first and ninety-ninth percentiles were -0.000730 and 0.000729. The mean fraction of exactly zero computed log returns was 0.007. These summaries described the starting markets without establishing stationarity or matching a calibrated empirical return distribution. The no-switch controls separately measured post-switch baseline financial variation across dealer depths (Supplementary Table S2).

The refined finite-size analysis resampled entire starting markets across fractions in 2,000 bootstrap replicates. The empirical estimate used the first upward crossing and allowed local decreases in probability, while the logistic fit imposed a smooth increasing response (Supplementary Table S11). Each replicate reselected the susceptibility maximum over the tested fractions. Consequently, intervals included variation in which sampled fraction supplied that maximum, but not interpolation bias, uncertainty from selecting the refinement grid, or changes from observing for longer. The recorded valid-bootstrap fraction identified samples in which the requested crossings remained bracketed. No susceptibility exponent or infinite-size limit was fitted.

A.6 Transaction-fee comparison and validation

The transaction-fee extension retained the original executed study and used a separate versioned simulation snapshot. Its manifest specified all 8,400 simulations before execution. The run checked cash conservation including the dealer's fee account, share conservation, nonnegative final trader resources, positive finite prices, the constant product of trading reserves, and the quoted price as their ratio. Dealer revenue was also checked against fees on every completed buy and sale. Each switched run shared its pre-switch financial and decision history with its no-switch control, including the first post-switch price and fee collection. With technical traders held, the complete Sell-fraction histories and final grids agreed exactly across fee rates and dealer depths. Separate tests checked zero-fee equivalence to the frozen original simulator, exhaustive feasible-quantity selection, fee-inclusive affordability, isolated round trips, balanced orders, and the zero-depth fixed-price case. Dealer fees were not available to fund trading in any condition. The additional source snapshot, manifest, per-run records, summaries, and validation report are stored in `output/rebuilt/transaction-fees-v1/`. The manuscript additions were generated from its validated summaries by `code/fees/WritePaper.py`. The fee study varied neither market size nor trader endowments and did not extend the observation period.

A.7 Revision timing and interior technical traders

The revision-timing and interior-technical-trader comparisons used a separate versioned study, `output/rebuilt/lazy-revision-interior-technical-v1/`, whose manifest fixed all 37,950 simulations in 1,650 groups before execution. Each group comprised one no-switch control and the switched runs at 22 fractions placed around the crossover found in a 20-market pilot of the same

condition. With revision probability $r = 1$ and no interior technical traders the simulation is the main model, and the 2,150 runs of this study that repeated a condition of the manuscript campaigns reproduced the archived final grids exactly. Every switched run shared its control's history through the first quoted price after the switch, and cash and share conservation held in every run, with a largest cash discrepancy of 3.1×10^{-6}. Whether a trader revised its intention at a step was decided by a counter-based draw from the simulation seed, step, and trader, so paired runs used identical draws and the generator used for initialization was unaffected. Interior technical positions were ranked by a counter-based score of the seed and position, which did not depend on the fraction f, and the selection of traders for the new rule skipped those positions. Midpoint intervals used 2,000 bootstrap samples of whole starting markets drawn once for all conditions, so differences between conditions sharing starting markets were paired. One condition, grid size $L = 128$ with interior fraction $\rho = 0.05$ and technical traders held, had three rather than five tested fractions with majority probabilities strictly between 0.05 and 0.95 (Supplementary Table S16).

A.8 Inventory information and observed intentions

The main model treated neighboring states as publicly communicated intentions whose execution was unverified. We tested a hypothetical information regime in which a trader knew whether a neighbor had any shares and read Hold instead of Sell when that neighbor had none. This comparison specified both the information available and the response to it, without assuming that actual market participants could observe one another's holdings. Let $q_j(t)$ be trader j's shares after settlement at step t, and let $s_j^{\mathrm{in}}(t)$ be the intention it would supply to an interior neighbor under the original update order. The observation mode was $v \in \{\mathrm{original}, \mathrm{inventory}\}$, and the signal read by a neighbor was $b_j^{(v)}(t)$, given by:

$$b_j^{(v)}(t) = \begin{cases} 1, & v = \mathrm{inventory},\ s_j^{\mathrm{in}}(t) = 2,\ q_j(t) = 0, \\ s_j^{\mathrm{in}}(t), & \text{otherwise.} \end{cases} \tag{12}$$

We replaced the inputs in the neighbor sum with these signals (Eq. 1). Hold contributed one to that sum; it did not remove the neighbor. Orders still used stored intentions and settled before the technical traders updated from price. We then constructed the visible signals and updated interior intentions together. The observation rule applied to interior and technical senders and changed no stored intention directly. Selling the last share therefore changed the signal read in the subsequent update, while acquiring a share allowed a later Sell intention to be read again. Buy signals were unchanged. Settlement and holdings could now change subsequent neighbor inputs even with technical traders held, providing a financial route into decisions that was absent under original observations. We tested one response to knowledge of inventory, without modeling a choice of announcements or strategic misreporting.

Each market used grid size $L = 64$, initial cash of 1,000 and ten shares per trader, and 1,000 baseline steps with original observations and Rule 19280. At step 1,001, four branches retained identical intentions, cash, shares, dealer reserves, and technical-trader state: original or inventory-conditioned observations, each with or without the rule switch. The new observation rule also began in its no-switch control at that step, so a change caused by the information regime alone was separated from its effect on the response to the switch. Every branch continued for 3,000 steps with reserve clearing, zero fees, and sales below purchase cost allowed. We used 50 new starting markets, with seeds 730001 to 730050, technical traders held or active, and dealer depths $h = 1$ and 100. The initial switched fractions were $f = 0.10$, 0.15, 0.175, 0.18, 0.20, and 0.25, with one no-switch control per observation mode, depth, technical-trader condition, and starting market, giving 2,800 trajectories. Before these runs, we fixed an extension rule for each depth and technical-trader condition: if inventory-conditioned majority probability was below 0.75 at fraction $f = 0.25$, add fractions $f = 0.35$, 0.50, 0.75, and 1.00 in both observation modes; if it was above 0.25 at $f = 0.10$, add $f = 0.025$ and 0.05. Extensions used the same starting markets and controls. Only this one extension round was allowed. A two-market pilot used different seeds and supplied no observations to the reported estimates. The primary setting and outcome were fixed before the main runs and were unaffected by the extension rule.

The primary outcome was the raw interior Sell fraction averaged over the last 500 steps, at fraction $f = 0.20$, dealer depth $h = 100$, and with technical traders held. Raw intentions were measured after the update, so a reduction in visible Sell signals alone could not count as reduced spreading. For starting market k, observation mode v, and switch indicator z (one for a switch, zero otherwise), the

outcome was Y_{kvz}. We defined the switch effect within each observation mode as δ_{kv} and its mean change between observation modes as Δ, over $n = 50$ starting markets, according to:

$$\delta_{kv} = Y_{kv1} - Y_{kv0}, \qquad \Delta = \frac{1}{n}\sum_{k=1}^{n} \left(\delta_{k,\text{inventory}} - \delta_{k,\text{original}}\right). \tag{13}$$

We reported all four branch means and a two-sided percentile 95% interval for Δ from 10,000 bootstrap samples of whole starting markets, retaining every matched branch and shared control. Secondary outcomes were reaching a raw Sell majority, maximum quoted-price drawdown including the quote before intervention, and completed interior sales per interior trader. Their intervals and all other comparisons were pointwise descriptions. Majority probabilities also had Wilson 95% intervals. We did not estimate a precise crossover midpoint from this coarse sweep or treat an interval containing zero as evidence of equivalence. The intervals at added fractions were conditional on the prescribed extension. For signal diagnostics, we divided the number of interior Sell inputs sent by traders with zero shares by all interior Sell inputs, counting each sender once per step and summing over the specified period. Input signals used holdings after settlement and were recorded before each update; raw Sell fractions were recorded afterward. Through-onset diagnostics included the inputs producing the first majority, or the full period if no majority formed. Histories for the first starting market's controls and fraction-$f = 0.20$ branches at every depth and technical-trader condition were selected before inspecting outcomes.

The study was archived in `output/rebuilt/inventory-visibility-v1/`, with hashes of the copied manuscript simulator, observation layer, analysis, manifests, and package environment. Tests checked exact original-mode recovery, unchanged signals when all sellers held shares, the last-share and reacquisition cases, technical-sender signals, and update timing. An original-mode full run reproduced the archived manuscript's final grid exactly, and the existing reserve-clearing and round-trip tests passed. All four branches shared identical pre-intervention state hashes and histories through the first post-intervention quoted price. Every run checked nonnegative holdings, cash and share conservation, positive prices, and the dealer's constant product. The extension added 1,600 trajectories, for 4,400 in total. The largest cash discrepancy was 6.78×10^{-7} cash units, shares were conserved exactly, and the largest relative change in the reserve product was 1.64×10^{-14}.

With technical traders held, dealer depth $h = 100$, and fraction $f = 0.20$, all 50 original switched markets reached a Sell majority, compared with none under inventory-conditioned observations; neither no-switch control condition produced a majority (Supplementary Table S17). The mean raw Sell fraction over the final 500 steps was 77.78% with original signals and 25.48% with inventory-conditioned signals, compared with 24.82% and 24.79% in their respective controls. The primary paired contrast, the change in the switch effect between observation modes, was -52.28 percentage points, with a 95% interval of [-54.46, -49.86]. In the original switched markets, zero-share traders supplied an average of 39.96% of interior Sell inputs through the first majority and 78.69% over the full observation period. The mean paired increase in drawdown over the no-switch control fell from 23.16 to 2.02 percentage points, while completed sales per interior trader rose from 490.11 to 748.22, so fewer selling intentions did not mean fewer completed sales (Supplementary Table S17). Extending the fraction sweep under the prespecified rule found no inventory-conditioned majority at fraction $f = 0.50$ in this setting, but 44 of 50 markets reached one at $f = 0.75$, where the late mean raw Sell fraction was 31.52% (Supplementary Fig. S2). The observation rule therefore changed both the response to a minority rule switch and the fractions at which majorities were observed, while a first majority remained distinct from a high late Sell fraction.

B Supplementary tables and figures

Table S1: Design of the four main comparisons. Active technical traders responded to the quoted price; held technical traders stayed in Hold. The grid-size comparison used active technical traders at dealer depth $h = 1$ and 100, and held technical traders at $h = 100$. Counts include the no-switch controls. Where a comparison tested several fractions, one control per starting market and condition served all of them. The cash-and-shares comparison had no no-switch controls.

Comparison	Grid sizes	Dealer h	Technical traders	Starting markets	Simulations
Rule switch	64	100	active	50	400
Dealer depth	64	0–100	active/held	50	9,600
Grid size	32–128	$1, 100$	active/held	50	9,750
Cash and shares	64	$0.1, 1, 100$	active/held	50	10,800

Table S2: Paired rule switches on a 64×64 grid over 3,000 post-switch steps. In A, majority probability p is the fraction of markets reaching an interior Sell fraction of at least one-half; median onset is conditional on reaching that value. Undefined means no run reached a majority. In B, DD is median maximum drawdown in percent. ΔDD is the median paired difference, in percentage points, with a pointwise 95% whole-market bootstrap confidence interval (CI). These contrasts separate the rule switch from baseline financial variation.

A. Rule switch at $h = 100$, active technical traders

Fraction f	Starting markets	No-switch p	Switched p	Median onset
0.10	50	0.00	0.00	Undefined
0.15	50	0.00	0.00	Undefined
0.20	50	0.00	1.00	1070.0
0.25	50	0.00	1.00	311.5

B. Dealer depth at $f = 0.20$, active technical traders

h	Starting markets	No-switch DD	Switched DD	Paired ΔDD [95% CI]
0	50	0.00	0.00	0.00 [0.00, 0.00]
0.1	50	77.04	99.99	22.96 [22.56, 23.74]
0.5	50	42.83	99.82	57.03 [56.25, 57.86]
1	50	32.92	99.52	66.55 [65.56, 67.71]
2	50	23.15	98.59	75.29 [74.78, 75.89]
5	50	14.03	94.43	80.31 [79.22, 80.96]
10	50	8.31	85.68	77.33 [76.71, 77.48]
100	50	0.75	25.15	24.39 [24.20, 24.47]

Table S3: Grid-size comparisons over 50 starting markets and 3,000 post-switch steps. Brackets give 95% confidence intervals (CI) from 2,000 whole-market bootstrap samples. n_{int} counts tested fractions with majority probability strictly between 0.05 and 0.95. The fractions f_{25}, f_{50}, and f_{75} are the first upward crossings of majority probability 0.25, 0.5, and 0.75, found by linear interpolation between tested fractions, and the width is $f_{75} - f_{25}$. The susceptibility maximum is taken over the tested fractions and is selected again in each bootstrap sample.

Technical traders	h	L	n_{int}	f_{50} [95% CI]	Width [95% CI]	Peak χ_L [95% CI]
Active	1	32	9	0.212 [0.208, 0.217]	0.020 [0.014, 0.028]	26.3 [22.2, 34.4]
Active	1	48	6	0.186 [0.184, 0.188]	0.011 [0.008, 0.019]	54.6 [41.3, 69.9]
Active	1	64	9	0.177 [0.175, 0.180]	0.008 [0.007, 0.013]	85.9 [79.9, 114.6]
Active	1	96	5	0.168 [0.167, 0.169]	0.005 [0.004, 0.007]	143.2 [127.4, 202.5]
Active	1	128	7	0.167 [0.165, 0.168]	0.007 [0.005, 0.009]	219.8 [198.5, 300.9]
Active	100	32	6	0.214 [0.210, 0.218]	0.020 [0.012, 0.028]	25.7 [22.2, 33.9]
Active	100	48	5	0.186 [0.184, 0.188]	0.011 [0.008, 0.022]	51.2 [38.6, 65.3]
Active	100	64	5	0.175 [0.174, 0.180]	0.011 [0.009, 0.014]	88.5 [72.2, 124.2]
Active	100	96	5	0.169 [0.165, 0.171]	0.009 [0.007, 0.011]	132.5 [119.7, 178.7]
Active	100	128	6	0.166 [0.165, 0.168]	0.007 [0.005, 0.009]	230.0 [199.8, 302.8]
Held	100	32	5	0.216 [0.211, 0.224]	0.023 [0.018, 0.027]	21.2 [16.5, 27.2]
Held	100	48	5	0.193 [0.184, 0.195]	0.019 [0.016, 0.022]	48.6 [40.3, 65.3]
Held	100	64	5	0.178 [0.176, 0.180]	0.015 [0.008, 0.019]	110.5 [78.5, 138.5]
Held	100	96	5	0.168 [0.167, 0.170]	0.009 [0.006, 0.012]	168.7 [137.6, 221.0]
Held	100	128	7	0.166 [0.165, 0.167]	0.008 [0.004, 0.010]	241.2 [202.6, 326.0]

Table S4: Probability of a Sell majority after changes in trader cash and shares over 50 starting markets and 3,000 post-switch steps. Each fraction includes 27 combinations of cash, shares, and dealer depth. With technical traders held, the probability was the same across those combinations; with technical traders active, it spanned the stated range.

Fraction f	Technical traders held	Technical traders active (range)
0.170	0.22	0.12–0.36
0.175	0.34	0.30–0.58
0.180	0.64	0.54–0.80
0.185	0.70	0.72–0.94

Table S5: Majority outcomes with technical traders active and held, paired by starting market, over 50 starting markets and 3,000 post-switch steps. Fractions 0.17 and 0.18 come from the unchanged-resource runs of the trader-resource comparison, which reproduced the dealer-depth runs exactly where the two overlapped. Only active and Only held count markets that reached a majority under one condition alone; held technical traders stayed in Hold from initialization, so paired markets need not share their state at the switch. The difference is active minus held majority probability, with a 95% interval (CI) from 2,000 bootstrap samples of whole starting markets. Average rows weight the listed fractions equally and count market–fraction pairs.

h	Fraction f	Active p	Held p	Only active	Only held	Difference [95% CI]
0.1	0.170	0.32	0.22	12	7	0.10 [-0.08, 0.26]
0.1	0.175	0.44	0.34	13	8	0.10 [-0.08, 0.28]
0.1	0.180	0.60	0.64	10	12	-0.04 [-0.22, 0.16]
0.1	0.185	0.80	0.70	8	3	0.10 [-0.02, 0.24]
0.1	0.190	0.98	0.90	5	1	0.08 [0.00, 0.18]
0.1	0.195	1.00	0.98	1	0	0.02 [0.00, 0.06]
0.1	0.175–0.195	Average		27	12	0.07 [0.02, 0.14]
0.1	0.170–0.195	Average		49	31	0.06 [0.00, 0.12]
1	0.170	0.20	0.22	7	8	-0.02 [-0.18, 0.12]
1	0.175	0.36	0.34	10	9	0.02 [-0.16, 0.20]
1	0.180	0.58	0.64	8	11	-0.06 [-0.22, 0.12]
1	0.185	0.80	0.70	11	6	0.10 [-0.06, 0.26]
1	0.190	0.96	0.90	4	1	0.06 [-0.02, 0.16]
1	0.195	0.98	0.98	1	1	0.00 [-0.06, 0.06]
1	0.175–0.195	Average		26	17	0.04 [-0.01, 0.10]
1	0.170–0.195	Average		41	36	0.02 [-0.04, 0.07]
100	0.170	0.12	0.22	3	8	-0.10 [-0.22, 0.02]
100	0.175	0.50	0.34	13	5	0.16 [0.00, 0.32]
100	0.180	0.60	0.64	12	14	-0.04 [-0.22, 0.16]
100	0.185	0.84	0.70	13	6	0.14 [-0.02, 0.30]
100	0.190	0.94	0.90	4	2	0.04 [-0.06, 0.14]
100	0.195	1.00	0.98	1	0	0.02 [0.00, 0.06]
100	0.175–0.195	Average		31	13	0.09 [0.02, 0.15]
100	0.170–0.195	Average		46	35	0.04 [-0.02, 0.10]

Table S6: A one-time instruction to sell versus a persistent rule switch at the same selected interior traders. Both conditions used dealer depth $h = 100$, 3,000 post-intervention steps, and no-switch controls. Entries are majority probabilities with 95% Wilson intervals.

Technical traders	f	Starting markets	Sell pulse	Persistent rule
Active	0.15	50	0.00 [0.00, 0.07]	0.00 [0.00, 0.07]
Active	0.20	50	0.00 [0.00, 0.07]	1.00 [0.93, 1.00]
Active	0.25	50	0.00 [0.00, 0.07]	1.00 [0.93, 1.00]
Held	0.15	50	0.00 [0.00, 0.07]	0.00 [0.00, 0.07]
Held	0.20	50	0.00 [0.00, 0.07]	1.00 [0.93, 1.00]
Held	0.25	50	0.00 [0.00, 0.07]	1.00 [0.93, 1.00]

Table S7: All single-response changes to Rule 19280 with dealer depth $h = 100$. Each entry gives majority probability with technical traders held / active over 20 starting markets and 3,000 post-switch steps. Outputs are Buy $= 0$, Hold $= 1$, and Sell $= 2$. This descriptive screen used no multiplicity adjustment.

Sum c	Output change	Rule	$f = 0.15$	$f = 0.20$	$f = 0.25$
0	$2 \to 0$	19278	0.00 / 0.00	0.00 / 0.00	0.00 / 0.00
0	$2 \to 1$	19279	0.00 / 0.00	0.00 / 0.00	0.00 / 0.00
1	$0 \to 1$	19283	0.00 / 0.00	0.00 / 0.00	0.00 / 0.00
1	$0 \to 2$	19286	0.00 / 0.00	1.00 / 1.00	1.00 / 1.00
2	$0 \to 1$	19289	0.00 / 0.00	0.00 / 0.00	0.00 / 0.00
2	$0 \to 2$	19298	0.95 / 0.95	1.00 / 1.00	1.00 / 1.00
3	$0 \to 1$	19307	1.00 / 1.00	1.00 / 1.00	1.00 / 1.00
3	$0 \to 2$	19334	1.00 / 1.00	1.00 / 1.00	1.00 / 1.00
4	$1 \to 0$	19199	0.00 / 0.00	0.00 / 0.00	0.00 / 0.00
4	$1 \to 2$	19361	1.00 / 1.00	1.00 / 1.00	1.00 / 1.00
5	$1 \to 0$	19037	0.00 / 0.00	0.00 / 0.00	0.00 / 0.00
5	$1 \to 2$	19523	1.00 / 1.00	1.00 / 1.00	1.00 / 1.00
6	$2 \to 0$	17822	0.00 / 0.00	0.00 / 0.00	0.00 / 0.00
6	$2 \to 1$	18551	0.00 / 0.00	0.00 / 0.00	0.00 / 0.00
7	$2 \to 0$	14906	0.00 / 0.00	0.00 / 0.00	0.00 / 0.00
7	$2 \to 1$	17093	0.00 / 0.00	0.00 / 0.00	0.00 / 0.00
8	$2 \to 0$	6158	0.00 / 0.00	0.00 / 0.00	0.00 / 0.00
8	$2 \to 1$	12719	0.00 / 0.00	0.00 / 0.00	0.00 / 0.00

Table S8: Adaptive versus frozen band widths for active technical traders at fraction $f = 0.175$ over 50 starting markets. Frozen widths kept σ_t at its value immediately before the rule switch, while the reference price μ_t continued to update. Entries p are Sell-majority probabilities with 95% Wilson confidence intervals (CI) in brackets. Sell is the mean fraction of technical traders in Sell over post-switch steps and starting markets. The probability at dealer depth $h = 0.1$ minus that at $h = 100$ was -0.06 with adaptive widths and -0.26 with frozen widths. Their paired difference was 0.20, with a 95% interval [-0.08, 0.50] from 2,000 whole-market bootstrap samples.

h	Adaptive p [95% CI]	Frozen p [95% CI]	Adaptive Sell	Frozen Sell
0	0.34 [0.22, 0.48]	0.34 [0.22, 0.48]	0.000	0.000
0.1	0.44 [0.31, 0.58]	0.22 [0.13, 0.35]	0.139	0.044
0.5	0.36 [0.24, 0.50]	0.38 [0.26, 0.52]	0.123	0.081
1	0.36 [0.24, 0.50]	0.40 [0.28, 0.54]	0.128	0.086
2	0.44 [0.31, 0.58]	0.40 [0.28, 0.54]	0.117	0.094
5	0.38 [0.26, 0.52]	0.36 [0.24, 0.50]	0.116	0.140
10	0.40 [0.28, 0.54]	0.38 [0.26, 0.52]	0.125	0.163
100	0.50 [0.37, 0.63]	0.48 [0.35, 0.61]	0.103	0.168

Table S9: Additional dealer-depth comparisons at fraction $f = 0.20$ with active technical traders and 50 starting markets over 3,000 post-switch steps. Because the dealer's cash and share reserves keep a constant product, the initial slope of the log quoted price against net purchases per trader is approximately $2/h$; the model has no separate price-impact coefficient. Depth changes funding and price sensitivity together. Drawdown is a percentage, and the dealer-rejection fraction uses post-switch eligible orders. Probability brackets are 95% Wilson intervals.

h	Initial slope $2/h$	Majority probability	Median drawdown (%)	Rejection
40	0.050	1.00 [0.93, 1.00]	48.62	1.41×10^{-6}
100	0.020	1.00 [0.93, 1.00]	25.15	6.04×10^{-7}
400	0.005	1.00 [0.93, 1.00]	7.08	2.82×10^{-7}

Table S10: Observation period and the fraction f_{50} at which half the markets reached a Sell majority, with technical traders held and dealer depth $h = 100$. Entries are interpolated f_{50} with 95% whole-market bootstrap intervals over 50 starting markets. Criterion columns give the interior Sell fraction a run had to reach to count as a majority. These runs used one fixed set of tested fractions rather than the adaptively refined set, so interpolated midpoints can differ slightly from those in Supplementary Table S3 for the same runs, for example 0.167 rather than 0.166 at $L = 128$ and 0.219 rather than 0.216 at $L = 32$.

L	Post-switch steps	Criterion 0.4	Criterion 0.5	Criterion 0.6
32	750	0.236 [0.228, 0.238]	0.249 [0.245, 0.255]	0.266 [0.260, 0.270]
32	1,500	0.216 [0.210, 0.220]	0.230 [0.228, 0.234]	0.245 [0.238, 0.250]
32	3,000	0.207 [0.205, 0.210]	0.219 [0.210, 0.225]	0.229 [0.226, 0.232]
64	3,000	0.172 [0.169, 0.176]	0.178 [0.176, 0.180]	0.186 [0.180, 0.188]
64	9,000	0.165 [0.163, 0.167]	0.167 [0.165, 0.168]	0.171 [0.169, 0.175]
128	3,000	0.163 [0.162, 0.164]	0.167 [0.165, 0.168]	0.169 [0.168, 0.171]
128	6,000	0.154 [0.153, 0.156]	0.157 [0.156, 0.158]	0.159 [0.158, 0.160]
128	12,000	0.150 [0.148, 0.151]	0.152 [0.150, 0.153]	0.153 [0.152, 0.155]

Table S11: Logistic fits and susceptibility maxima over 50 starting markets and 3,000 post-switch steps. Parentheses give whole-market bootstrap standard errors (SE). The fitted curve imposes increasing probability; interpolated crossings do not. The final column gives the tested fraction with the largest susceptibility, so its precision is limited by the spacing of nearby tested fractions.

Technical traders	h	L	Logistic f_{50} (SE)	Logistic width (SE)	f at peak
Active	1	32	0.212 (0.0015)	0.020 (0.0017)	0.217
Active	1	48	0.185 (0.0013)	0.014 (0.0012)	0.190
Active	1	64	0.177 (0.0006)	0.010 (0.0007)	0.185
Active	1	96	0.168 (0.0005)	0.006 (0.0006)	0.175
Active	1	128	0.166 (0.0005)	0.006 (0.0005)	0.168
Active	100	32	0.215 (0.0015)	0.018 (0.0017)	0.217
Active	100	48	0.185 (0.0011)	0.013 (0.0014)	0.190
Active	100	64	0.177 (0.0008)	0.009 (0.0007)	0.190
Active	100	96	0.168 (0.0006)	0.007 (0.0005)	0.175
Active	100	128	0.166 (0.0004)	0.005 (0.0004)	0.168
Held	100	32	0.218 (0.0016)	0.018 (0.0014)	0.225
Held	100	48	0.189 (0.0011)	0.013 (0.0009)	0.190
Held	100	64	0.178 (0.0010)	0.011 (0.0008)	0.185
Held	100	96	0.169 (0.0006)	0.007 (0.0006)	0.172
Held	100	128	0.166 (0.0005)	0.006 (0.0004)	0.168

Table S12: The five example markets of Fig. 4 and all 50 starting markets at fraction $f = 0.15$ with their no-switch controls, extended from 3,000 to 10,000 steps with dealer depth $h = 100$ and active technical traders. Times are steps after the rule switch, following 1,000 baseline steps. Drawdowns and price changes are percentages; price changes use the quoted price at the rule switch as reference. Final Sell averages the last 500 steps of the longer observation period. None means no Sell majority by 10,000 steps. The examples were selected from the 3,000-step summaries before any images were inspected. The extensions follow the same starting markets, and are not independent replications.

A. Five examples selected from new 3,000-step results

f	Starting market	First majority	Drawdown 3,000 steps	Drawdown 10,000 steps	Price change 10,000 steps	Final Sell
0.150	2	None	5.22	5.22	-3.29	0.268
0.175	15	4,073	10.17	26.34	-26.32	0.897
0.175	41	2,330	22.14	25.51	-25.49	0.594
0.185	5	1,836	23.77	26.07	-26.03	0.910
0.200	38	1,028	25.61	26.50	-26.48	0.913

B. All 50 starting markets at $f = 0.150$

Decision rule	Steps	Majorities / 50	Median drawdown	Worst drawdown	Median price change
New rule at 15%	3,000	0/50	5.04	12.29	-3.40
New rule at 15%	5,000	0/50	5.25	17.65	-3.49
New rule at 15%	10,000	2/50	5.87	25.30	-3.54
Original rule	3,000	0/50	0.75	1.15	0.06
Original rule	5,000	0/50	0.81	1.15	0.04
Original rule	10,000	0/50	0.88	1.15	0.04

Table S13: Transaction-fee comparisons. Each condition used 50 starting markets. The fee was charged to each completed buyer and seller from initialization. Sell-majority probabilities are followed by Wilson 95% intervals. In Panel A, drawdown and cumulative dealer fees are medians for the switched runs at fraction $f = 0.20$. Drawdown uses the 3,000-step post-switch observation period, including the quoted price immediately before the switch. Fees include all 4,000 trading steps and are expressed as a percentage of initial aggregate trader cash. Fee income was held separately from trading reserves. Panel B reports paired differences between 25 and zero basis points with percentile 95% intervals from 2,000 whole-market bootstrap samples. The probability difference is a difference in proportions; drawdown differences are median paired changes in percentage points. The complete protocol, all six tested fractions, and paired differences are archived with the transaction-fee study.

A. Outcomes at each fee rate

h	Technical	Fee (bp)	Majority, $f = 0.175$	Majority, $f = 0.20$	Drawdown (%)	Fees (%)
0.1	held	0	0.34 [0.22, 0.48]	1.00 [0.93, 1.00]	99.99	0.00
0.1	held	5	0.34 [0.22, 0.48]	1.00 [0.93, 1.00]	99.99	14.19
0.1	held	10	0.34 [0.22, 0.48]	1.00 [0.93, 1.00]	99.99	26.25
0.1	held	25	0.34 [0.22, 0.48]	1.00 [0.93, 1.00]	99.98	52.85
0.1	active	0	0.44 [0.31, 0.58]	1.00 [0.93, 1.00]	99.99	0.00
0.1	active	5	0.38 [0.26, 0.52]	1.00 [0.93, 1.00]	99.99	13.48
0.1	active	10	0.30 [0.19, 0.44]	1.00 [0.93, 1.00]	99.98	25.73
0.1	active	25	0.44 [0.31, 0.58]	1.00 [0.93, 1.00]	99.98	52.90
1	held	0	0.34 [0.22, 0.48]	1.00 [0.93, 1.00]	99.53	0.00
1	held	5	0.34 [0.22, 0.48]	1.00 [0.93, 1.00]	99.49	13.01
1	held	10	0.34 [0.22, 0.48]	1.00 [0.93, 1.00]	99.45	24.32
1	held	25	0.34 [0.22, 0.48]	1.00 [0.93, 1.00]	99.32	50.30
1	active	0	0.36 [0.24, 0.50]	1.00 [0.93, 1.00]	99.52	0.00
1	active	5	0.44 [0.31, 0.58]	1.00 [0.93, 1.00]	99.50	12.19
1	active	10	0.46 [0.33, 0.60]	0.98 [0.90, 1.00]	99.48	23.83
1	active	25	0.40 [0.28, 0.54]	1.00 [0.93, 1.00]	99.40	49.26
100	held	0	0.34 [0.22, 0.48]	1.00 [0.93, 1.00]	24.66	0.00
100	held	5	0.34 [0.22, 0.48]	1.00 [0.93, 1.00]	24.27	7.12
100	held	10	0.34 [0.22, 0.48]	1.00 [0.93, 1.00]	23.80	14.15
100	held	25	0.34 [0.22, 0.48]	1.00 [0.93, 1.00]	23.21	34.91
100	active	0	0.50 [0.37, 0.63]	1.00 [0.93, 1.00]	25.15	0.00
100	active	5	0.40 [0.28, 0.54]	1.00 [0.93, 1.00]	24.94	7.29
100	active	10	0.34 [0.22, 0.48]	1.00 [0.93, 1.00]	24.51	14.45
100	active	25	0.40 [0.28, 0.54]	1.00 [0.93, 1.00]	23.67	36.19

B. Selected paired comparisons with technical traders active

h	f	Outcome	0 bp	25 bp	Paired difference [95% CI]
100	0.175	Majority probability	0.50	0.40	-0.10 [-0.28, 0.08]
100	0.2	Drawdown (%)	25.15	23.67	-1.34 [-1.60, -1.16]
1	0.175	Drawdown (%)	76.52	92.81	9.04 [3.46, 16.98]

Table S14: Following experiments on 64×64 grids in which every interior trader kept Rule 19280, with 1,000 baseline steps and 3,000 steps of following. Followers were interior traders chosen at random. Cash is the mean over followers of each follower's final cash, after every trader sold all of its shares to the dealer, divided by its cash and shares at the quoted price when following began, minus the same mean for the same traders in the paired no-following control. Non-follower Sell is the change in the fraction of time that non-followers spent in Sell, and drawdown is the change in the maximum drawdown of the quoted price. All changes are in percentage points, with pointwise 95% intervals from resampling whole starting markets. In A, each starting market had two random assignments of followers and advice. In the masked condition, neighbors read each follower's intention from the no-following control, and the cash difference is advice minus masked. The two endpoints fixed before these markets were simulated, cash with advice and the cash difference at dealer depth $h = 100$ with technical traders held, had Bonferroni-adjusted 97.5% intervals of [25.43, 28.06] and [22.30, 25.30]. In the no-following controls at $h = 100$ with technical traders held, the same non-followers spent 24.9% of their time in Sell. In B, the change in the final 500 steps after advice stopped exceeded 20 percentage points in 38 of 100 runs and was below 5 percentage points in 61. In C, copying repeated a randomly chosen trader's completed trade from the previous step.

A. Shared advice from one source to 40% of interior traders, 50 starting markets

h	Technical	Cash, advice	Cash, masked	Cash difference	Non-follower Sell	Drawdown
100	held	26.71 [25.58, 27.88]	2.89 [2.18, 3.58]	23.82 [22.52, 25.10]	19.12 [18.61, 19.63]	27.53 [27.46, 27.60]
100	active	20.14 [19.05, 21.25]	-2.56 [-3.25, -1.88]	22.70 [21.48, 23.92]	18.89 [18.39, 19.40]	27.35 [27.25, 27.44]
1	held	-3.12 [-4.42, -1.81]	-12.52 [-12.83, -12.21]	9.40 [8.02, 10.86]	19.12 [18.61, 19.63]	55.35 [54.84, 55.85]
1	active	-18.08 [-18.32, -17.85]	-12.71 [-12.91, -12.51]	-5.37 [-5.49, -5.25]	18.81 [18.30, 19.33]	67.04 [66.42, 67.63]

B. Stopping advice and separate advice, 40% followers, $h = 100$, technical traders held

Advice	Cash	Non-follower Sell, final 500 steps	Non-follower Sell	Drawdown
One source	26.71 [25.58, 27.88]	19.35 [18.20, 20.57]	19.12 [18.61, 19.63]	27.53 [27.46, 27.60]
Stopped at 1,000 steps	12.19 [11.35, 12.99]	22.83 [16.71, 29.20]	21.61 [17.49, 25.91]	27.28 [27.19, 27.37]
Separate for each follower	4.87 [4.84, 4.89]	-13.00 [-13.05, -12.95]	-13.00 [-13.03, -12.96]	4.34 [4.28, 4.41]

C. Exploratory comparison, 10 starting markets, 40% followers, $h = 100$, technical traders held

Followers	Cash	Non-follower Sell	Drawdown
Advice, one source	27.01 [24.03, 30.00]	18.90 [18.07, 19.73]	27.56 [27.38, 27.73]
Advice, sixteen sources	6.13 [5.68, 6.63]	-10.92 [-11.10, -10.71]	11.70 [11.35, 12.02]
Advice, separate for each	4.89 [4.83, 4.94]	-13.01 [-13.13, -12.90]	4.36 [4.23, 4.50]
Copying one trader	0.58 [-0.04, 1.30]	4.07 [-6.56, 16.03]	9.22 [3.85, 15.30]
Copying sixteen traders	0.81 [0.66, 0.97]	-18.31 [-18.70, -17.87]	0.28 [-0.33, 0.93]

Table S15: Revision timing. At each step, each interior trader adopted its rule's output with revision probability r and otherwise kept its current intention; $r = 1$ is the main model. Entries are the interpolated fraction f_{50} at which half of 50 starting markets reached a Sell majority within 3,000 post-switch steps, with 95% intervals in brackets from 2,000 bootstrap samples of whole starting markets, drawn once and shared by all conditions. Active − held is the difference in f_{50} at dealer depth $h = 100$, from the same samples. Halves is the largest median, over the three tested fractions nearest the held f_{50}, of the absolute difference between the Sell fractions of the two checkerboard halves of the grid, averaged over five snapshots in the last 400 steps. Panel B uses technical traders held at dealer depth $h = 100$ and revision probability $r = 0.9$; the width is $f_{75} - f_{25}$. No no-switch control reached a Sell majority in any condition.

A. Grid size $L = 64$

r	Held	Active, $h = 1$	Active, $h = 100$	Active − held, $h = 100$	Halves
1	0.178 [0.176, 0.180]	0.178 [0.175, 0.181]	0.175 [0.174, 0.180]	−0.003 [−0.005, 0.003]	0.254
0.95	0.253 [0.250, 0.257]	0.256 [0.250, 0.258]	0.255 [0.253, 0.258]	0.002 [−0.002, 0.006]	0.014
0.9	0.298 [0.295, 0.301]	0.296 [0.290, 0.300]	0.298 [0.295, 0.301]	0.000 [−0.004, 0.003]	0.011
0.75	0.358 [0.357, 0.361]	0.358 [0.354, 0.361]	0.360 [0.350, 0.362]	0.001 [−0.009, 0.004]	0.011
0.5	0.416 [0.404, 0.419]	0.413 [0.405, 0.417]	0.412 [0.409, 0.415]	−0.004 [−0.007, 0.009]	0.009

B. Revision probability $r = 0.9$, technical traders held

L	Held	Width
32	0.334 [0.331, 0.346]	0.037 [0.028, 0.045]
64	0.298 [0.295, 0.301]	0.017 [0.012, 0.031]
128	0.285 [0.283, 0.286]	0.008 [0.006, 0.014]

Table S16: Technical traders inside the grid, with revision probability $r = 1$. In addition to the technical traders along the edge, band-rule technical traders occupied a fraction ρ of interior positions, excluding the traders given the new rule; the row of 63 positions adds as many interior technical traders as there are along the edge. Reach is the mean fraction of the other interior traders with at least one technical trader among their four neighbors. Entries are interpolated f_{50} over 50 starting markets and 3,000 post-switch steps, with 95% intervals in brackets from 2,000 bootstrap samples of whole starting markets shared by all conditions; active − held is the difference in f_{50} at dealer depth $h = 100$. Panel B uses interior fraction $\rho = 0.05$. At grid size $L = 128$ with technical traders held, majority probability rose steeply enough that only three tested fractions had probabilities strictly between 0.05 and 0.95. No no-switch control reached a Sell majority in any condition.

A. Grid size $L = 64$

Interior	Reach	Held	Active, $h = 1$	Active, $h = 100$	Active − held, $h = 100$
None	1.6%	0.178 [0.176, 0.180]	0.178 [0.175, 0.181]	0.175 [0.174, 0.180]	−0.003 [−0.005, 0.003]
63 positions	7.8%	0.187 [0.184, 0.191]	0.219 [0.217, 0.221]	0.224 [0.222, 0.227]	0.037 [0.032, 0.042]
2.5%	11.0%	0.191 [0.188, 0.193]	0.244 [0.240, 0.247]	0.254 [0.252, 0.258]	0.063 [0.061, 0.067]
5%	19.6%	0.200 [0.195, 0.203]	0.320 [0.309, 0.328]	0.349 [0.345, 0.355]	0.149 [0.145, 0.156]
10%	35.1%	0.240 [0.238, 0.243]	0.385 [0.365, 0.397]	0.368 [0.334, 0.404]	0.128 [0.089, 0.164]

B. Interior fraction $\rho = 0.05$

L	Held	Active, $h = 100$	Active − held
32	0.223 [0.219, 0.228]	0.367 [0.363, 0.380]	0.144 [0.137, 0.158]
64	0.200 [0.195, 0.203]	0.349 [0.345, 0.355]	0.149 [0.145, 0.156]
128	0.191 [0.189, 0.192]	0.337 [0.334, 0.340]	0.146 [0.144, 0.149]

Table S17: Inventory-conditioned signals at fraction $f = 0.20$ over 50 new starting markets and 3,000 post-intervention steps. Original observations passed every intention; inventory-conditioned observations passed Hold instead of Sell from a sender with no shares. Each mode had a switched branch and a no-switch control with identical pre-intervention histories. Entries are branch means, except majority entries, which count the markets reaching a raw Sell fraction of one-half. Late Sell fraction averages the final 500 steps. The final column is the inventory-conditioned switch effect minus the original switch effect, with a percentile 95% interval from 10,000 bootstrap samples of whole starting markets (Eq. 13). Differences in Panels A–C are percentage points; Panel D uses completed sales per interior trader. The primary outcome was Panel A with technical traders held and dealer depth $h = 100$; other comparisons are descriptive.

		Original signals		Inventory-conditioned		
h	Technical	Control	Switch	Control	Switch	Difference [95% interval]
A. Late raw Sell fraction (%)						
1	held	24.82	77.78	24.46	24.96	-52.46 [-54.63, -50.03]
1	active	25.03	80.96	24.69	25.11	-55.51 [-57.57, -53.11]
100	held	24.82	77.78	24.79	25.48	-52.28 [-54.46, -49.86]
100	active	24.99	79.92	24.95	25.63	-54.25 [-56.54, -51.67]
B. Markets reaching a Sell majority (out of 50)						
1	held	0	50	0	0	-100.00 [-100.00, -100.00]
1	active	0	49	0	0	-98.00 [-100.00, -94.00]
100	held	0	50	0	0	-100.00 [-100.00, -100.00]
100	active	0	50	0	0	-100.00 [-100.00, -100.00]
C. Mean maximum drawdown (%)						
1	held	44.39	99.44	42.31	42.84	-54.51 [-55.65, -53.34]
1	active	33.04	99.36	31.76	33.35	-64.72 [-65.96, -63.49]
100	held	1.20	24.36	1.22	3.23	-21.14 [-21.37, -20.88]
100	active	0.73	24.93	0.68	3.01	-21.87 [-22.11, -21.58]
D. Mean completed sales per interior trader						
1	held	735.29	490.06	731.80	726.89	240.32 [210.98, 267.73]
1	active	741.87	494.84	738.67	732.28	240.65 [214.05, 266.74]
100	held	743.97	490.11	743.55	748.22	258.53 [228.85, 286.27]
100	active	749.11	489.87	747.95	753.07	264.36 [231.47, 296.41]

In the primary switched condition with original observations, traders with no shares supplied 78.69% of all interior Sell inputs after the switch and 39.96% through the first majority (averages over starting markets). Inputs were measured after settlement and before the decision update. The full fraction sweep for this condition is shown in Supplementary Fig. S2.

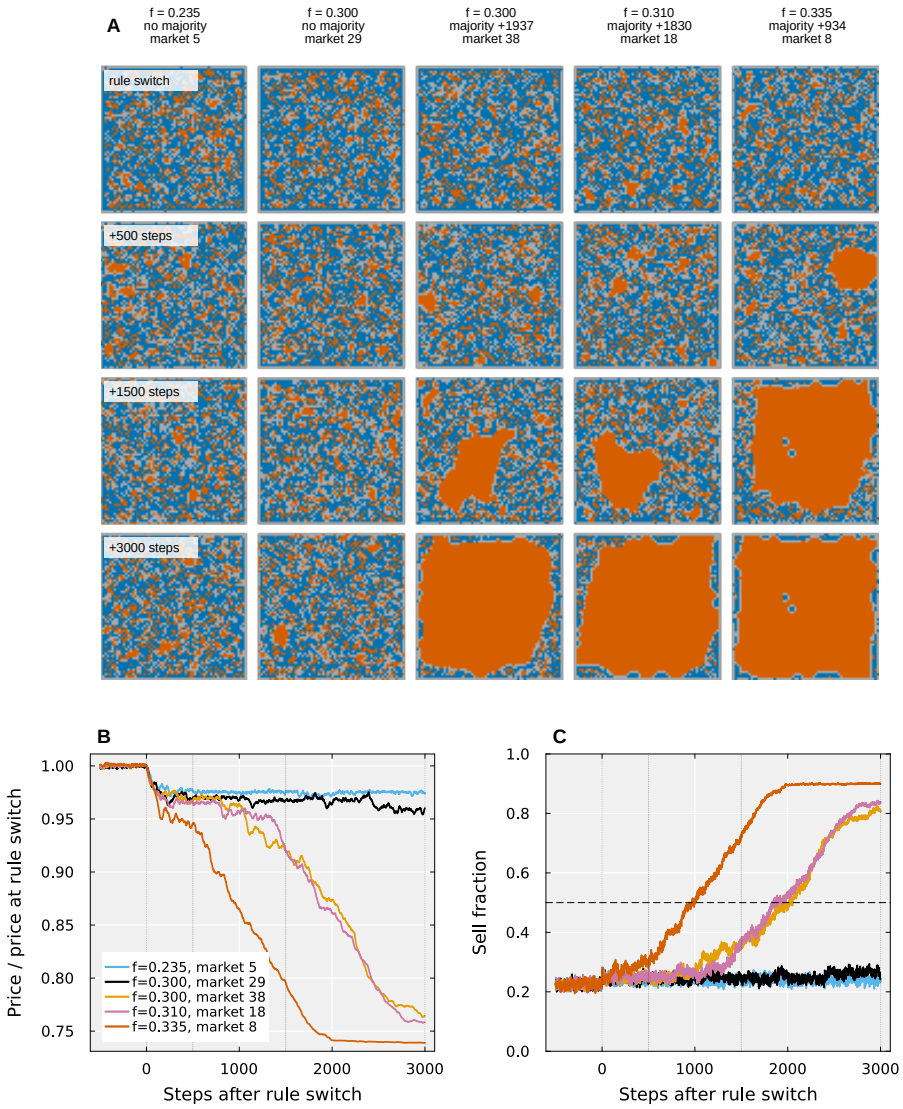


Figure S1: Five example markets with revision probability $r = 0.9$, active technical traders, and dealer depth $h = 100$ over 3,000 post-switch steps, the counterpart of Fig. 4 without the separate checkerboard halves of the main model. Blue, gray, and orange denote Buy, Hold, and Sell. The fractions $f = 0.235$, 0.300, 0.310, and 0.335 were the tested fractions whose majority probabilities at revision probability $r = 0.9$ (0.00, 0.62, 0.84, and 1.00) were nearest those of the fractions in Fig. 4 at $r = 1$ (0.00, 0.50, 0.84, and 1.00), with ties broken by proximity to the $r = 0.9$ midpoint. Within each fraction and outcome, the market nearest the median onset, or the median final Sell fraction for markets without a majority, was selected from the campaign summaries before any images were inspected, with the starting-market number breaking ties. **(A)** Grid decisions at the rule switch and after 500, 1,500, and 3,000 steps. Starting markets were 5, 29, 38, 18, and 8, from left to right. **(B)** Quoted price divided by its value at the rule switch. **(C)** Interior Sell fraction, with one-half marked by a horizontal line. Midpoints and widths appear in Supplementary Table S15.

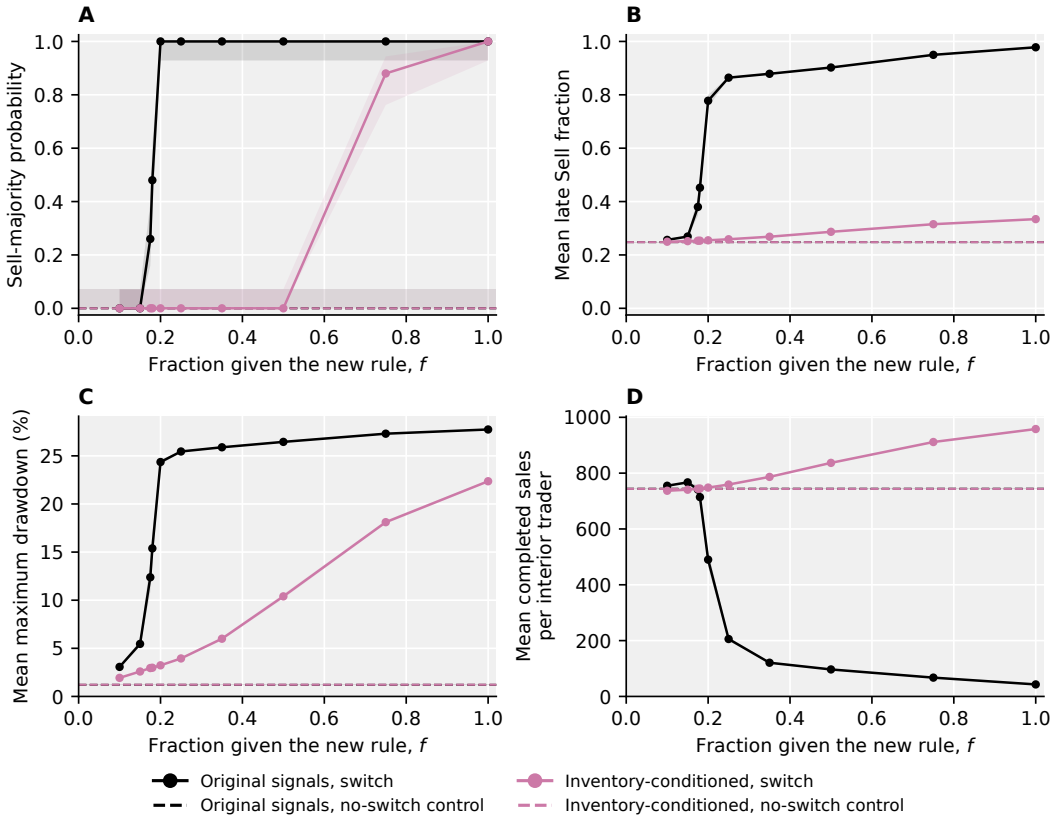


Figure S2: Inventory information in the primary setting: technical traders held, dealer depth $h = 100$, and grid size $L = 64$. Black denotes original signals and purple inventory-conditioned signals. Solid lines and points denote switched markets; dashed horizontal lines denote the shared no-switch controls. Each point used the same 50 starting markets and 3,000 post-intervention steps; joining lines guide the eye. **(A)** Probability of reaching a raw Sell majority, with Wilson 95% intervals. **(B)** Mean raw Sell fraction over the final 500 steps. **(C)** Mean maximum quoted-price drawdown. **(D)** Mean completed sales per interior trader. Bands in B–D are pointwise percentile 95% intervals from 10,000 bootstrap samples of whole starting markets. The original sweep covered fractions $f = 0.10$ to 0.25; larger fractions were added under the prespecified extension rule (Appendix A.8). Controls retain Rule 19280 in both observation modes, so the change in control outcomes measures the information intervention alone. The comparison tests one hypothetical response to observing inventory.